

Axonal distribution of mitochondria maintains neuronal autophagy during aging via eIF2 β

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In flies defective for axonal transport of mitochondria, the authors report the upregulation of one subunit, the beta subunit, of the heterotrimeric eIF2 complex via mass spectroscopy proteomics. Neuronal overexpression of eIF2 β phenocopied aspects of neuronal dysfunction observed when axonal transport of mitochondria was compromised. Conversely, lowering eIF2 β expression suppressed aspects of neuronal dysfunction. While these are intriguing and **useful** observations, technical weaknesses limit the interpretation. On balance, the evidence supporting the current claims is suggestive but **incomplete**, especially concerning the characterization of the eIF2 heterotrimer and the data regarding translational regulation.

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Abstract

Summary

Neuronal aging and neurodegenerative diseases are accompanied by proteostasis collapse, while the cellular factors that trigger it have not been identified. Impaired mitochondrial transport in the axon is another feature of aging and neurodegenerative diseases. Using *Drosophila*, we found that genetic depletion of axonal mitochondria causes dysregulation of protein degradation. Axons with mitochondrial depletion showed abnormal protein accumulation and autophagic defects. Lowering neuronal ATP levels by blocking glycolysis did not reduce autophagy, suggesting that autophagic defects are associated with mitochondrial distribution. We found that eIF2 β was increased by the depletion of axonal mitochondria via proteome analysis. Phosphorylation of eIF2 α , another subunit of eIF2, was lowered, and global translation was suppressed. Neuronal overexpression of eIF2 β phenocopied the autophagic defects and neuronal dysfunctions, and lowering eIF2 β

expression rescued those perturbations caused by depletion of axonal mitochondria. These results indicate the mitochondria-eIF2 β axis maintains proteostasis in the axon, of which disruption may underlie the onset and progression of age-related neurodegenerative diseases.

Highlights

- Loss of axonal mitochondria impairs autophagy and accumulates proteins in the axon
- Loss of axonal mitochondria increases eIF2 β and decreases p-eIF2 α
- Neuronal upregulation of eIF2 β induces autophagic defects and locomotor dysfunction
- Lowering eIF2 β rescues autophagic defects caused by loss of axonal mitochondria

Introduction

Neurons have a morphologically complex architecture composed of microcompartments and require tight regulation of the abundance of proteins and organelles spatially and temporally¹. Such control of protein amounts, or proteostasis, is essential for neuronal functions² and is achieved through the orchestration of protein expression, folding, trafficking, and degradation controlled by intrinsic and environmental signals³. Translation is initiated by the eukaryotic initiation factor 2 (eIF2) complex⁴. eIF2, a heterotrimer of α , β , and γ subunits, transports Met-tRNA to the ribosome in a GTP-dependent manner⁵. Under stressed conditions, phosphorylation of eIF2 α attenuates global translation and initiates translation of mRNAs related to the integrated stress response (ISR)⁶. As for protein degradation, autophagy and proteasome are major systems that maintain proteostasis⁷. The proteasome degrades unnecessary proteins, followed by regulated ubiquitination processes⁸, and autophagy removes damaged or harmful components, including large protein aggregates and organelles, through catabolism (selective autophagy)⁹. In addition to autophagy induced by acute stressors, a basal level of selective autophagy mediates the global turnover of damaged proteins¹⁰.

Such constitutive autophagy decreases during aging, which may underlie declines in the structural and functional integrity of neurons¹¹. Decreased protein degradation and accumulation of abnormal proteins also contribute to increased risks of neurodegenerative diseases. Age-related neurodegenerative diseases such as Alzheimer's disease and Parkinson's disease are often associated with the accumulation of misfolded proteins such as amyloid- β , tau, and α -synuclein¹². Enhancement of autophagy mitigates age-related dysfunctions and neurodegeneration caused by proteotoxic stress¹³. However, it is not fully understood how aging disrupts the regulation of this constitutive autophagy.

Neurons are also highly energy-demanding cells. At nerve terminals, action potentials trigger the release of neurotransmitters via exocytosis of synaptic vesicles, which requires a constant supply of ATP and calcium buffering¹⁴. Such neuronal activity relies on mitochondrial functions¹⁵, and mitochondria are actively transported from their major sites of biogenesis in soma to axons¹⁶. However, the axonal transport of mitochondria declines during aging^{17,18,19}. Reduced axonal transport of mitochondria is thought to contribute to age-related declines in neuronal functions^{17,19,20,21}. The number of functional mitochondria in synapses is reduced in the brains of patients suffering from age-related neurodegenerative diseases such as Alzheimer's disease²², and mutations in genes involved in mitochondrial dynamics are linked to neurodegenerative diseases²³. The mislocalization of mitochondria is sufficient to cause age-dependent neurodegeneration in *Drosophila* and mice^{24,25}, indicating that the proper

distribution of mitochondria is essential to maintain neuronal functions. Thus, depletion of functional mitochondria from axons and proteostasis collapse are common features of aging and neurodegenerative diseases.

Mitochondrial transport is regulated by a series of molecular adaptors that mediate the attachment of mitochondria to molecular motors¹⁶. In *Drosophila*, mitochondrial transport is mediated by *milton* and *Miro*, which attaches mitochondria to microtubules via kinesin heavy chain^{26,27}. In the absence of *milton* or *Miro*, synaptic terminals and axons lack mitochondria, although mitochondria are numerous in the neuronal cell body²⁸. We previously reported that RNAi-mediated knockdown of *milton* or *Miro* in neurons causes a reduction in axonal mitochondria, age-dependent locomotor defects²⁹, and age-dependent neurodegeneration in neuropile area starting around 30 days after eclosion (day-old)²⁴, and enhances axon degeneration caused by human tau proteins²⁴, suggesting that these flies can be used as a model to analyze the effect of depletion of axonal mitochondria during aging. In this study, we investigated a causal relationship between mitochondrial distribution and neuronal proteostasis by using neuronal knockdown of *milton*. We found that depletion of axonal mitochondria reduced autophagy and increased the accumulation of aggregated proteins in the axon prior to gross neurodegeneration. Proteome analysis and follow-up biochemical analyses revealed that neuronal knockdown of *milton* increased eIF2 β levels and lowered phosphorylation of eIF2 α in the axon. In addition, *milton* knockdown suppressed global translation. Overexpression of eIF2 β was sufficient to decrease autophagy and induce neuronal dysfunction, and genetic suppression of eIF2 β restored autophagy and improved neuronal function in the *milton* knockdown background. These findings suggest that loss of axonal mitochondria and elevated levels of eIF2 β mediate proteostasis collapse and neuronal dysfunction during aging.

Results

Depletion of axonal mitochondria by knockdown of *milton* or *Miro* causes protein accumulation in the axon

In *Drosophila*, mitochondrial transport is mediated by *milton* and *Miro*, which attach mitochondria to microtubules via kinesin heavy chain^{26,27} (Figure 1A). It has been reported that expression of *milton* RNAi in neurons via pan-neuronal elav-GAL4 driver reduced *milton* protein levels in *Drosophila* head lysate to 40% and mito-GFP signals in axons to 50%^{24,29}.

To test how loss of axonal mitochondria affects proteostasis in neurons, we first examined the accumulation of ubiquitinated proteins. At 14-day-old, more ubiquitinated proteins were deposited in the brains of *milton* knockdown flies than in those of age-matched control flies (Figure 1B, $p < 0.005$ between control RNAi and *milton* RNAi). There was no significant increase in ubiquitinated proteins in *milton* knockdown flies at 1-day old, suggesting that the accumulation of ubiquitinated proteins caused by *milton* knockdown is age-dependent (Figure S1). We also analyzed the effect of the neuronal knockdown of *Miro*, a partner of *milton*, on the accumulation of ubiquitin-positive proteins. Since severe knockdown of *Miro* in neurons causes lethality, we used UAS-*Miro* RNAi strain with low knockdown efficiency, whose expression driven by elav-GAL4 caused 30% reduction of *Miro* mRNA in head extract²⁴. Although there was a tendency for increased ubiquitin-positive puncta in *Miro* knockdown brains, the difference was not significant (Figure 1B, $p > 0.05$ between control RNAi and *Miro* RNAi). These data suggest that the depletion of axonal mitochondria induced by *milton* knockdown leads to the accumulation of ubiquitinated proteins before neurodegeneration occurs.

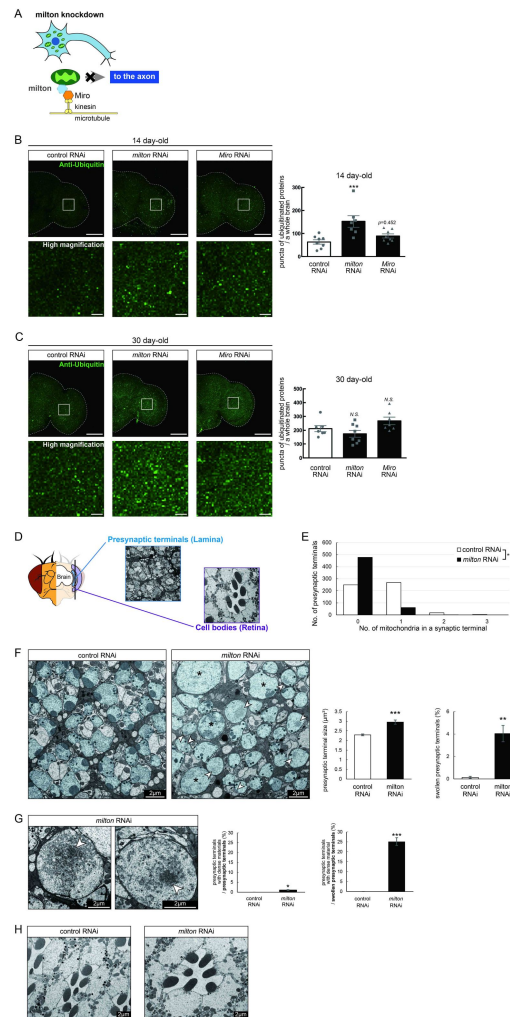


Figure 1.

Knockdown of *milton* or *Miro* causes protein accumulation in the axon

(A) Schematic representation of the mitochondrial transport machinery. Knockdown of *milton*, an adapter protein for mitochondrial transport, depletes mitochondria in the axon. (B, C) Ubiquitinated proteins in brains with neuronal knockdown of *milton* or *Miro*. Brains dissected at 14-day-old (B) or 30-day-old (C) were immunostained with an antibody against ubiquitin. Firefly *luciferase* RNAi was used as a control. Representative images (left) and quantitation of the number of ubiquitin-positive puncta (right) are shown. Scale bars of hemibrains, 100 μ m, Scale bars of high magnifications, 10 μ m. Means $\pm$ SE, $n = 8$. *N.S.*, $p > 0.05$; $***p < 0.005$ (one-way analysis of variance (ANOVA) followed by Tukey's honestly significant difference (HSD) *post hoc* test). (D) Cross sections in the lamina and in the retina were used to analyze the ultrastructure of synapses and cell bodies, respectively. *milton* RNAi was expressed in the retina and neurons via a combination of GAL4 drivers, a pan-retinal *gmr*-GAL4 and pan-neuronal *elav*-GAL4. (E) Quantitation of the number of mitochondria in a presynaptic terminal from transmission electron micrographs. 180 presynaptic terminals from cross-sections of the lamina from three brains were analyzed. $***p < 0.005$ (Chi-square test). (F, G) Presynaptic terminals of photoreceptor neurons of control and *milton* knockdown flies. Photoreceptor neurons are highlighted in blue. Swollen presynaptic terminals (asterisks in (F)), characterized by the enlargement and higher circularity, were found more frequently in *milton* knockdown neurons. Arrowheads indicate presynaptic terminals with dense materials. Scale bars, 2 μ m. Representative images (Left) and quantitation (Right) are shown. 918–1118 from three heads were quantified for the percentage of swollen presynaptic terminals, and 180 presynaptic terminals from three heads were quantified for the size of presynaptic terminals. Mean $\pm$ SE, $**p < 0.01$, $***p < 0.005$ (Student's *t*-test). (G) Dense materials (arrowheads in (G)) in the presynaptic terminals of *milton* knockdown neurons. Scale bars, 2 μ m. The ratio of presynaptic terminals containing dense materials were quantified from 918–1118 presynaptic terminals from three heads. Mean $\pm$ SE, $***p < 0.005$ (Student's *t*-test). (H) Cell bodies of photoreceptor neurons of control and *milton* knockdown flies. Scale bars, 2 μ m. Flies were 27-day-old.

It has been reported that ubiquitinated proteins accumulate with aging³⁰; thus, we analyzed the accumulation of ubiquitinated proteins in aged brains (30-day-old) with *milton* knockdown. The number of puncta of ubiquitinated proteins did not significantly differ between control and *milton* knockdown flies or between control and *Miro* knockdown flies (**Figure 1C**, $p > 0.05$). These results suggest that depletion of axonal mitochondria may have more impact on proteostasis in young neurons than in old neurons.

We examined the ultrastructure of presynaptic terminals and cell bodies in photoreceptor neurons with *milton* knockdown by transmission electron microscopy in 27-day-old flies (**Figure 1D**). As previously reported²⁴, the number of mitochondria in presynaptic terminals decreased in *milton* knockdown (**Figure 1E**). The swelling of presynaptic terminals, characterized by the enlargement and roundness, was not reported at 3-day-old²⁴ but observed at this age with about 4% of total presynaptic terminals (**Figure 1F**, asterisks).

Some presynaptic terminals of *milton* knockdown neurons contained dense materials (**Figure 1F and G**, arrowheads). Dense materials are rarely found in age-matched control neurons, indicating that *milton* knockdown induces abnormal protein accumulation in the presynaptic terminals (**Figure 1G and H**). In *milton* knockdown neurons, dense materials are found in swollen presynaptic terminals more often than in presynaptic terminals without swelling, suggesting a positive correlation between the disruption of proteostasis and axonal damage (**Figure 1G**). In contrast, dense materials were not observed in cell bodies in the *milton* knockdown retina (**Figure 1H**). These results indicate that the depletion of axonal mitochondria induces protein accumulation in the axon.

Depletion of axonal mitochondria impairs protein degradation pathways

Since abnormal proteins were accumulated in *milton* knockdown brains, we next examined if protein degradation pathways were suppressed. We analyzed autophagy via western blotting of the autophagy markers LC3 and p62³¹. During autophagy progression, LC3 is conjugated with phosphatidylethanolamine to form LC3-II, which localizes to isolation membranes and autophagosomes. LC3-I accumulation occurs when autophagosome formation is impaired, and LC3-II accumulation is associated with lysosomal defects^{31,32}. p62 is an autophagy substrate, and its accumulation suggests autophagic defects^{31,32}. We found that *milton* knockdown increased LC3-I, and the LC3-II/LC3-I ratio was lower in *milton* knockdown flies than in control flies at 14-day-old (**Figure 2A**). We also analyzed p62 levels in head lysates sequentially extracted using detergents with different stringencies (1% Triton X-100 and 2% SDS). Western blotting revealed that p62 levels were increased in the brains of 14-day-old of *milton* knockdown flies (**Figure 2B**). The increase in the p62 level was significant in the Triton X-100-soluble fraction but not in the SDS-soluble fraction (**Figure 2B**), suggesting that depletion of axonal mitochondria impairs the degradation of less-aggregated proteins. Proteasome activity was also significantly decreased in brains with neuronal knockdown of *milton* (**Figure 2C**, $p < 0.005$).

At 30-day-old, LC3-I was still higher, and the LC3-II/LC3-I ratio was lower, in *milton* knockdown compared to the control (**Figure 2D**). At this age, *milton* knockdown increased p62 significantly in 1% Triton X-100 fraction and 2% SDS fraction (**Figure 2E**). Proteasome activities were also decreased in *milton* knockdown flies at 30-day-old (**Figure 2F**). These results indicate that depletion of axonal mitochondria impairs protein degradation pathways.

ATP deprivation does not impair autophagy

milton knockdown downregulates ATP in the axon³³. To examine whether the disruption of protein degradation pathways by *milton* knockdown is due to ATP deprivation, we investigated the effects of knocking down phosphofructokinase (*Pfk*), a rate-limiting enzyme in glycolysis, on protein degradation pathways. Neuronal knockdown of *Pfk* was reported to lower ATP levels in

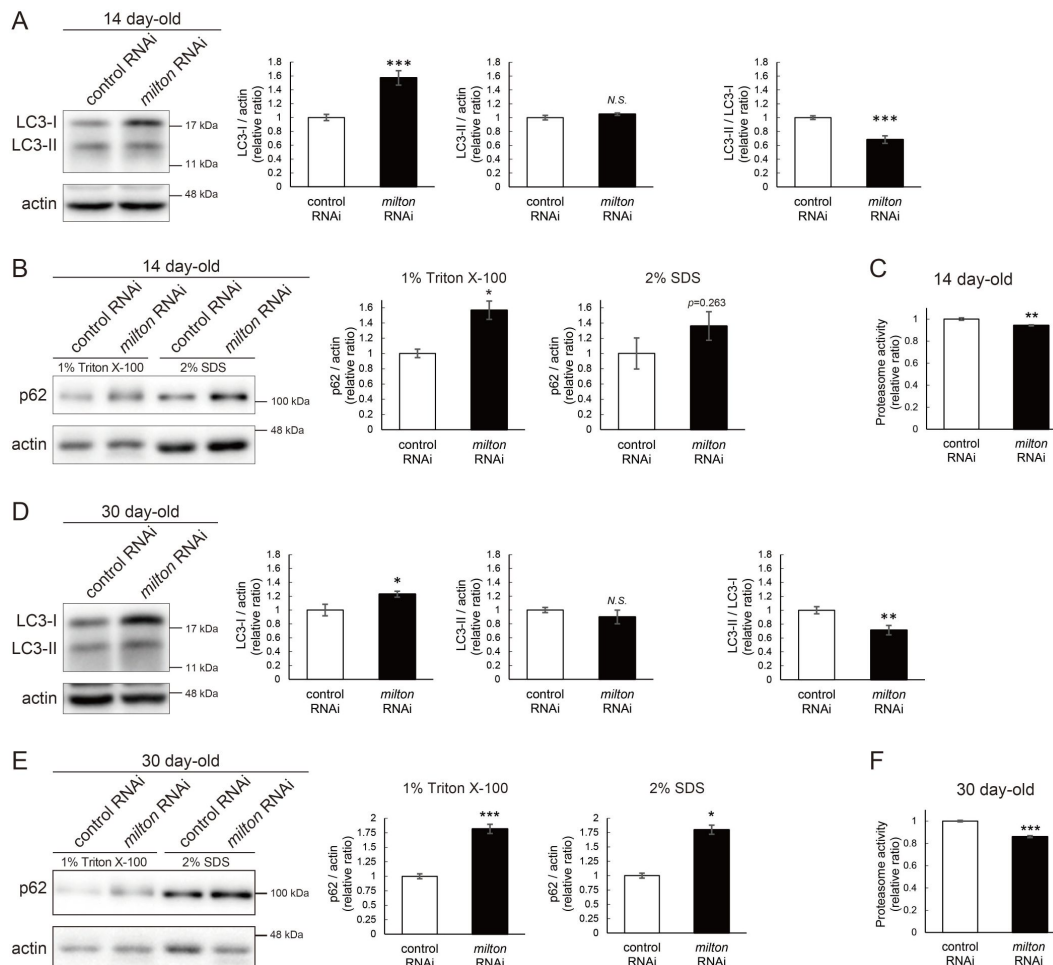


Figure 2.

***milton* knockdown impairs protein degradation pathways**

(A, B) Western blotting of head extracts of control and *milton* knockdown flies with antibodies against LC3 (A) and Ref2P, the fly homolog of mammalian p62 (B). For the analyses of p62 levels, heads were extracted with 1% Triton X-100 or 2% SDS (B). Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 6$ (LC3), $n = 3$ (p62). (C) Proteasome activity in head extracts of control and *milton* knockdown flies was measured by hydrolysis of Suc-LLVY-AMC at 14-day-old. Means $\pm$ SE, $n = 3$. (D, E) Western blotting of head extracts of 30-day-old control and *milton* knockdown flies. Blotting was performed with anti-LC3 (D) and anti-p62 (E) antibodies. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 6$ (LC3), $n = 3$ (p62). (F) Proteasome activity in head extracts of 30-day-old control and *milton* knockdown flies. Means $\pm$ SE, $n = 3$. N.S., $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$ (Student's *t*-test).

brain neurons³³. *Pfk* knockdown and *milton* knockdown decreased ATP to similar levels (**Figure 3A-C**). However, in contrast with *milton* knockdown, *Pfk* knockdown did not affect the levels of LC3-I, LC3-II or the LC3-II/LC3-I ratio (**Figure 3D**). *Pfk* knockdown decreased p62 level (**Figure 3E**), suggesting that autophagy is promoted. On the other hand, proteasome activity was decreased by *Pfk* knockdown (**Figure 3F**). These results suggest that the downregulation of axonal ATP upon depletion of axonal mitochondria decreases proteasome activity, but not autophagy.

Proteome analysis suggests that depletion of axonal mitochondria causes disruption of autophagy and premature aging

To identify the pathways that mediate the decrease in autophagy in *milton* knockdown brains, we performed proteome analysis to systematically detect differentially expressed proteins upon neuronal knockdown of *milton*. We analyzed flies at 7- and 21-day-old, the age before autophagic defects are detected and the age just before the onset of neurodegeneration, respectively (**Figure 4A**). 1039 proteins were detected by liquid chromatography-tandem mass spectrometry (LC-MS/MS). Expression of 36 proteins was significantly increased (22 proteins) or decreased (14 proteins) by *milton* knockdown at 7-day-old (**Figure 4B**, **Tables 1** and S1). At 21-day-old, the expression of 41 proteins (31 upregulated and ten downregulated proteins) was significantly altered in *milton* knockdown flies compared with control flies (**Figure 4C**, **Tables 1** and S1). The “Interaction search” algorithm using KeyMolnet showed that proteins whose expression was significantly altered in the brains of *milton* knockdown flies at both 7- and 21-day-old were closely associated with the autophagic pathway (**Table 2**). Proteins involved in pathways characteristics of aging, such as the immune response (Transcriptional regulation by STAT), cancer (Transcriptional regulation by SMAD, Transcriptional regulation by myc), longevity (Transcriptional regulation by FOXO, Sirtuin signaling pathway), and stress responses (HSP90 signaling pathway, MAPK signaling pathway)^{34,35}, were enriched in the proteome profiles of *milton* knockdown flies compared with those of control flies at 7-day-old (**Table 2**), suggesting that depletion of axonal mitochondria accelerates aging in the brain.

7 day-old				21 day-old			
Accession ^a	Name	Abundance Ratio: (7 days, <i>milton</i> KD) (7 days, control)	Abundance Ratio P-Value: (7 days, <i>milton</i> KD) (7 days, control)	Accession ^a	Name	Abundance Ratio: (21 days, <i>milton</i> KD) (21 days, Control)	Abundance Ratio P-Value: (21 days, <i>milton</i> KD) (21 days, Control)
G9V751	Atacin-B	100	1E-17	G10714	Angiotensin-converting enzyme	100	1E-17
G04448	Bifunctional methyltetrahydrofolate dehydrogenase/cyclohydrolase, mitochondrial	100	1E-17	G0H028	Cecropin-A2	100	1E-17
P22700	Calcium-transporting ATPase sarcoplasmic/endoplasmic reticulum type	100	1E-17	G9V958	Cytochrome P450 4p1	100	1E-17
G9V056	Cytochrome P450 4p1	100	1E-17	P51592	G3.15-subunit-protein, lysine hyd	100	1E-17
P11052	FAMF1-related protein	100	1E-17	G9V917	Lysine-specific demethylase 4d	100	1E-17
P09611	Formin 1 of Myosin heavy chain, muscle	100	1E-17	G9V934	Heat-activating factor and hydrolyase III subunit beta homolog	100	1E-17
G9V051	Probable cytochrome P450 12A5, mitochondrial	100	1E-17	G9V930	Probable 26S ribosomal protein S25, mitochondrial	100	1E-17
G17020	Toll-like receptor 7	100	1E-17	G9V951	Probable phosphotyrosine kinase regulatory subunit alpha	100	1E-17
G9A900	Oxidation-reduction protein COQ9, mitochondrial	100	1E-17	G9V932	Protein arginase-2	100	1E-17
G9V120	Naa-Pro aminopeptidase Apep	100	1E-17	P54359	Septin-2	100	1E-17
G9V918	Bannan, bicoid	4.038	1E-17	G24402	Septin-A	15.716	1E-17
P07701	Salivary glue protein Sgs-5	2.843	1.86025E-09	G9V933	Glycogen-binding subunit 78A	8.886	1E-17
G17052	Pectin hydrolase domain-containing family F member 1 homolog	2.538	2.22045E-09	G17052	Phenylalanyl-tRNA synthetase subunit SB1	6.659	1E-17
P81041	Alpha-amylase B	2.684	7.44847E-09	G9V931	Carboxyl-terminal protein caspase	6.626	1E-17
P11051	Formin 4 of Troponin T, skeletal muscle	2.86	6.65635E-11	P81041	Alpha-amylase B	5.722	1E-17
G9V128	Mitochondrial import inner membrane translocase subunit TIM44	2.91	3.54025E-01	P48804	Atacin-A	4.567	1E-17
P41375	Eukaryotic translation initiation factor 2 subunit 2	2.465	3.38486E-09	G9V938	Histone H3.3A	4.778	1E-17
G9V933	Sequestosome 1B1	2.465	2.25279E-09	P26625	Protein sort of sequestosome	4.658	1E-17
R72399	Protein channel Oligo	2.382	1.71741E-06	P02515	Heat shock protein 22	4.479	1E-17
G9V925	DNA polymerase II transcriptional coactivator	2.445	6.70245E-09	G9V946	Atacin-C	4.450	1E-17
G9V944	Protein, Lysine 5	2.127	8.38665E-13	G17911	Subunit A of Potassium voltage-gated channel protein Shal	4.160	1E-17
P27779	Plasmin activator Eps-70E	2.113	1.00215E-09	G9V930	Activity-regulated cytoskeleton associated protein 1	3.271	1E-17
G9V918	Probable G2E-Luciferase synthase	0.468	2.12015E-08	P14109	Protein red p2	3.014	1E-17
P55035	26S proteasome non-ATPase regulatory subunit 4	0.487	3.50158E-11	G9V932	Probable small nuclear ribonucleoprotein Sm D1	2.43	4.91672E-13
G9V946	26S ribosomal protein L19, mitochondrial	0.487	6.00551E-13	G9V944	Subunit A of RNA polymerase subunit 2 homolog A	2.289	2.6564E-11
G9V902	Cytosolic Fe-S cluster assembly factor HUBP2 homolog	0.468	6.34514E-10	G9V935	Nonmen Bicp1a1	2.24	5.16671E-11
G9V940	Probable maleylacetoacetate isomerase 1	0.432	4.50595E-09	P22939	Heat shock protein 67B1	2.223	7.90048E-11
G9V932	Probable pseudouridine-C5-phosphatase	0.419	1.09021E-14	P27779	Heat shock protein 67B1	2.192	1.65444E-10
G9V908	Cytochrome P450 4p2	0.398	4.11328E-10	G9V933	Serine/threonine-protein kinase tricomered	2.059	4.15071E-09
G14385	Lysosomal protein 2	0.373	1.32271E-14	G9V937	Chitinase-like protein 5d2	2.005	4.61162E-09
G9V946	Lysosomal thioesterase PPT2 homolog	0.369	2.55255E-09	G9V931	Atacin-B	2.038	6.71416E-09
G24114	Division abnormally delayed protein	0.301	2.24466E-09	G9V935	Probable 3-hydroxyacyl-CoA dehydrogenase, mitochondrial	0.491	7.42652E-09
G9V946	Atacin-C	0.301	1E-17	P54359	ATP synthase protein 6	0.421	1.858E-15
P26993	Inositol 1,4,5-trisphosphate receptor	0.01	1E-17	P33438	Glucanin	0.414	8.13731E-13
G9V933	Open rectifier potassium channel protein 1	0.01	1E-17	G9V944	Protein Tumorid C	0.218	1E-17
G9V915	UNC95-like protein	0.01	1E-17	G9V943	Protein Tumorid C	0.195	1E-17
				G9V939	CAMP-specific 3'-5'-cyclic phosphodiesterase	0.01	1E-17
				G9V935	Open rectifier potassium channel protein 1	0.01	1E-17
				G9V933	Probable maleylacetoacetate isomerase 1	0.01	1E-17
				G9V940	Protein draper	0.01	1E-17
				A12710	Serine/threonine-protein kinase N	0.01	1E-17

^a UniProt accession number

Depletion of axonal mitochondria upregulates eIF2β and decreases phosphorylation of eIF2α

Differentially expressed proteins at 7-day-old flies may reflect alterations that are causal for autophagic defects. We noticed that the expression level of eIF2β was 2.465-fold higher in the brains of *milton* knockdown flies than in those of control flies (**Figure 4B and D**). Upregulation of eIF2β in the brains of *milton* knockdown flies was confirmed by western blotting. *milton* knockdown increased eIF2β protein levels more than twice (**Figure 4E**), but did not change the level of eIF2β mRNA (**Figure 4F**).

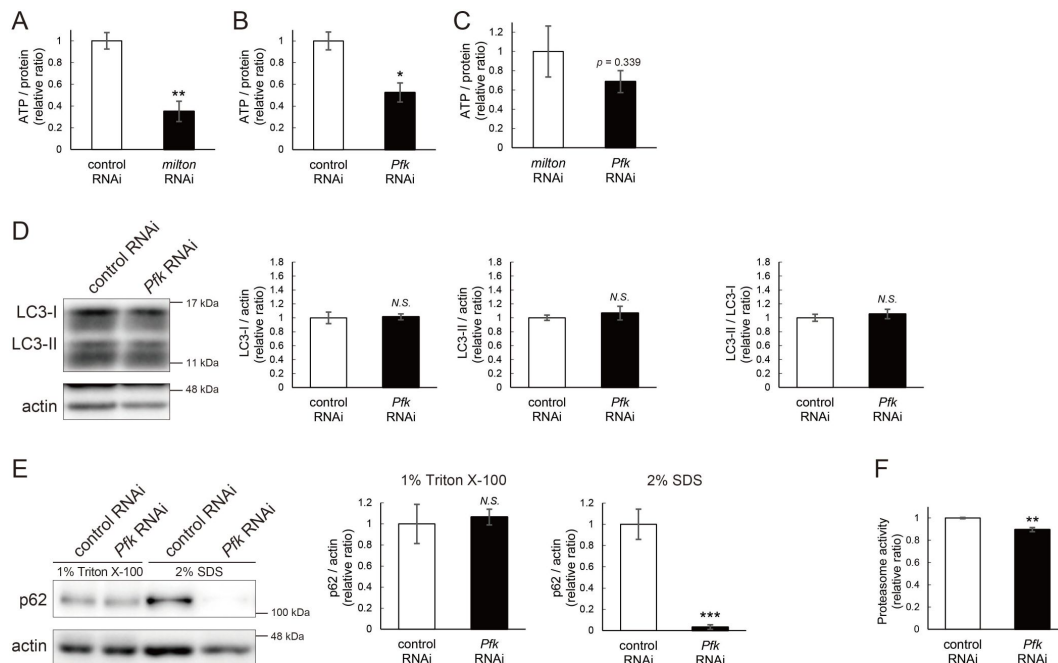


Figure 3.

ATP deprivation does not impair autophagy

(A-C) ATP levels in brain extracts of control and *milton* knockdown flies (A) and control and *Pfk* knockdown flies (B) and comparison of the effects of *milton* knockdown and *Pfk* knockdown on ATP levels (C). Flies were 14-day-old. Means $\pm$ SE, $n = 3$. (D, E) Western blotting of head extracts of flies with neuronal expression of control or *Pfk* RNAi. Blotting was performed with anti-LC3 (D) and anti-p62 (E) antibodies. For analyses of p62 levels, heads were extracted with 1% Triton X-100 or 2% SDS. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 6$ (LC3), $n = 3$ (p62). (F) Proteasome activity in head lysates of flies with neuronal expression of control or *Pfk* RNAi was measured by hydrolysis of Suc-LLVY-AMC. Means $\pm$ SE, $n = 3$. N.S., $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$ (Student's t -test). Flies were at 14-day-old.

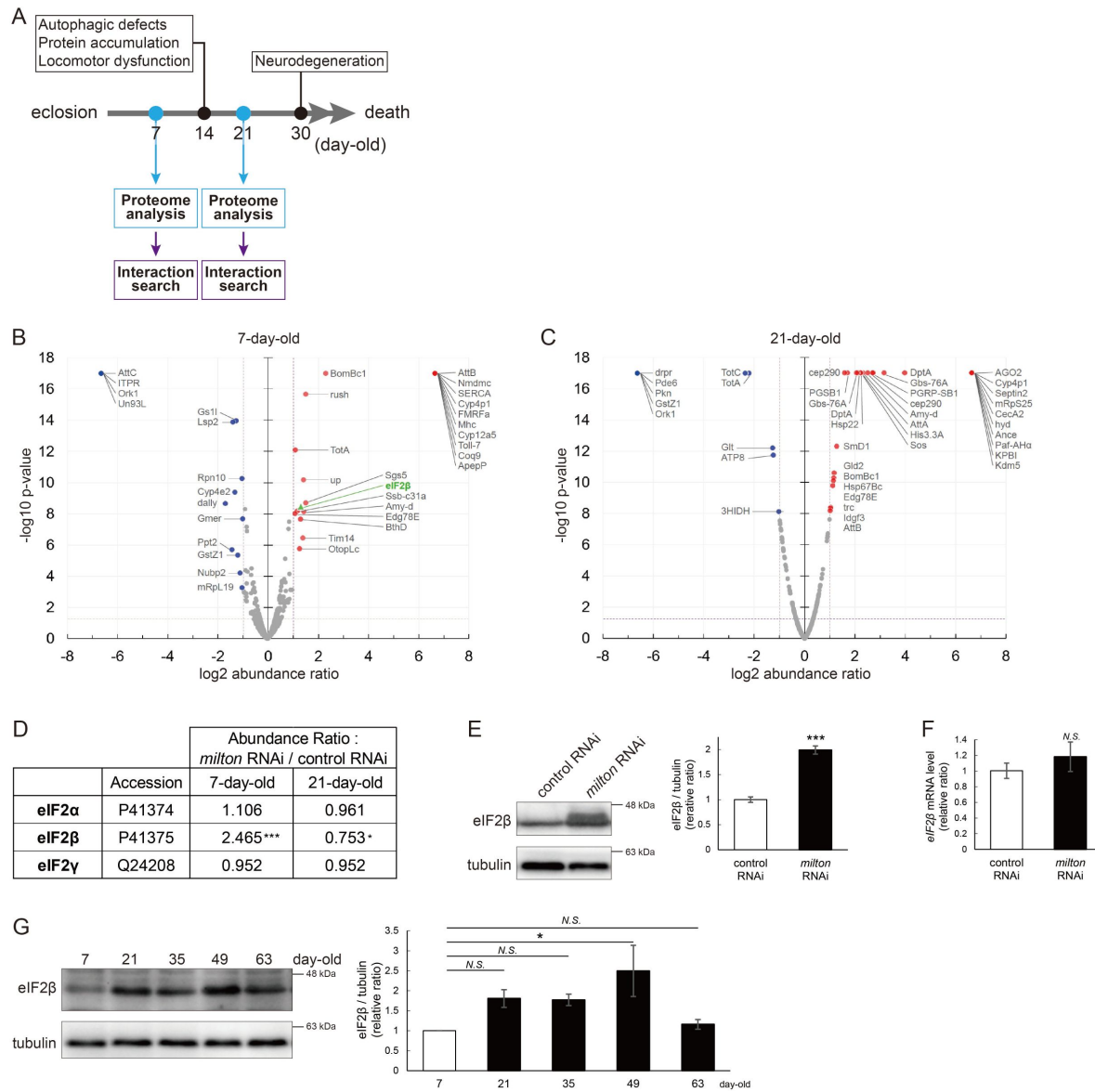


Figure 4.

milton knockdown upregulates eIF2 β in young flies

(A) Timing of proteome analysis and phenotypes observed in *milton* knockdown flies. (B, C) Scatter plots of the $\log_2$ abundance ratio (x-axis) against the $-\log_{10} p\text{-value}$ (y-axis) of proteins at 7-day-old (B) and 21-day-old (C). (D) eIF2 subunit protein levels from proteome analysis of *milton* knockdown flies compared to those of control flies. (E) Western blotting of head extracts of flies expressing control or *milton* RNAi in neurons with an anti-eIF2 β antibody. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 6$. (F) eIF2 β mRNA levels quantified by qRT-PCR. Means $\pm$ SE, $n = 4$. (G) Western blotting of head extracts of wild-type flies with an anti-eIF2 β antibody. Flies were 7-, 21-, 35-, 49- and 63-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 3$, * $p < 0.05$ (one-way analysis of variance (ANOVA) followed by Dunnett's multiple comparison test).

We also investigated age-dependent changes in eIF2 β by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. eIF2 β levels increased during aging until 49-day-old (**Figure 4G**). These results suggest that upregulation of eIF2 β in *milton* knockdown fly brain reflects early an onset of age-dependent increase of eIF2 β levels. eIF2 β is a subunit of the eukaryotic initiation factor 2 (eIF2) complex, which is critical for translation initiation and the integrated stress response (ISR)⁴. eIF2 is a heterotrimer of α , β , and γ subunits, and eIF2 α is phosphorylated during the ISR⁶. As for the other subunits of eIF2 complex, proteome analysis did not detect a significant difference in the protein levels of eIF2 α and eIF2 γ between *milton* knockdown and control flies at 7- and 21-day-old (**Figure 4D**). Western blotting of brain lysates showed that *milton* knockdown reduced eIF2 α levels (**Figure 5A**), while p-eIF2 α levels were not significantly affected (**Figure 5B**).

To analyze local changes of eIF2 α and p-eIF2 α , we carried out immunostaining. We focused on the mushroom body, where axons, dendrites, and cell bodies can be easily identified (**Figure 5C**). Both eIF2 α and p-eIF2 α were downregulated in the cell body (Kenyon cells) and dendritic (Calyxes) regions of the brains of *milton* knockdown flies (**Figure 5D**). In axons (lobe tips), *milton* knockdown did not affect eIF2 α (**Figure 5E**, $p = 0.271$) but significantly downregulated p-eIF2 α (**Figure 5E**). The ratio of p-eIF2 α to eIF2 α was lower in the axon but not in the soma or dendritic region. These results suggest that axonal distribution of mitochondria regulate the level of overall eIF2 α protein and local p-eIF2 α .

Depletion of axonal mitochondria suppressed global translation

Phosphorylation of eIF2 α induces conformational changes in the eIF2 complex and inhibits global translation³⁶. To analyze the effects of *milton* knockdown on translation, we performed polysome gradient centrifugation to examine the level of ribosome binding to mRNA. Since p-eIF2 α was downregulated, we hypothesized that *milton* knockdown would enhance translation. However, unexpectedly, we found that *milton* knockdown significantly reduced the level of mRNAs associated with polysomes (**Figure 6A and B**). We also compared the level of translation between the brains of control and *milton* knockdown flies by assessing the incorporation of puromycin (**Figure 6C**). Puromycin incorporation was lower in the brains of *milton* knockdown flies than in those of control flies, while it was not statistically significant (**Figure 6C**, indicated by a bracket). These data suggest that the depletion of axonal mitochondria suppresses global translation.

eIF2 β upregulation reduces the level of p-eIF2 α , impairs autophagy, and decreases locomotor function

We were motivated to ask if eIF2 β upregulation mediates autophagic defects caused by *milton* knockdown. If so, neuronal overexpression of *eIF2 β* would also induce autophagy impairment. Neuronal overexpression of *eIF2 β* increased LC3-II, while the LC3-II/LC3-I ratio was not significantly different (**Figure 7A and B**). Overexpression of *eIF2 β* significantly increased the p62 level in the Triton X-100-soluble fraction (**Figure 7C**, 4-fold vs. control, $p < 0.005$ (1% Triton X-100)) but not in the SDS-soluble fraction (**Figure 7C**, 2-fold vs. control, $p = 0.062$ (2% SDS)), as observed in brains of *milton* knockdown flies (**Figure 2B**). These data suggest that neuronal overexpression of *eIF2 β* accumulates autophagic substrates.

Since *milton* knockdown reduced the p-eIF2 α level (**Figure 5E**), we asked whether an increase in eIF2 β affects p-eIF2 α . Neuronal overexpression of *eIF2 β* did not affect the eIF2 α level but significantly decreased the p-eIF2 α level (**Figure 7D and E**).

Depletion of axonal mitochondria causes age-dependent decline in locomotor function²⁴. We found that neuronal overexpression of *eIF2 β* also caused locomotor dysfunction (**Figure 7F**). Locomotor functions were significantly impaired in those flies at 20-day-old and worsened further

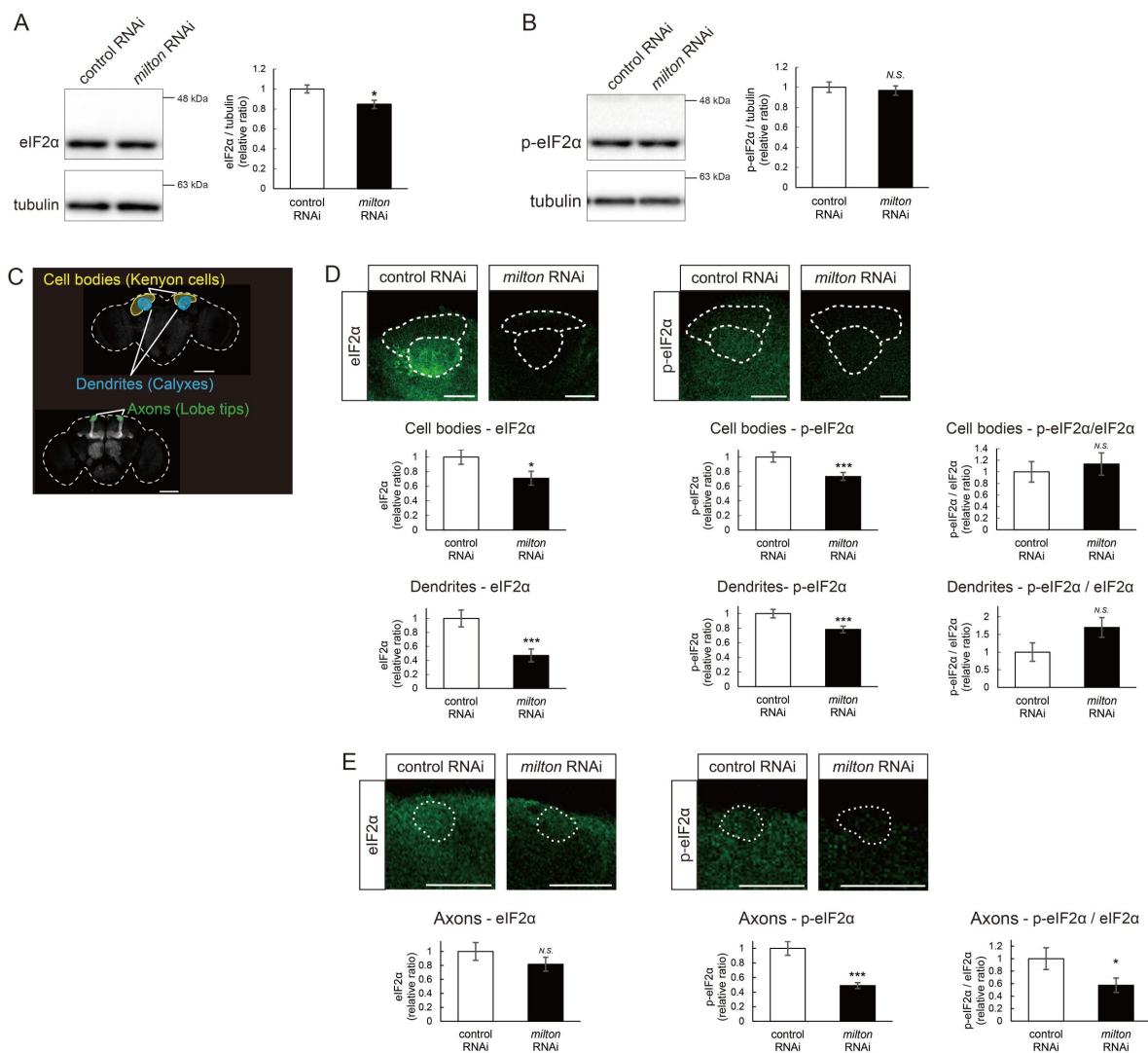


Figure 5.

***milton* knockdown decreases phosphorylation of eIF2α**

(A, B) Western blotting of head extracts with anti-eIF2α (A) and anti-p-eIF2α (B) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 6$. (C) A schematic representation of the axon (Lobe tips), the cell body region (Kenyon cells), and dendritic region (Calyxes) in the fly brain. Scale bars, 100 μ m. (D, E) Immunostaining with anti-eIF2α and anti-p-eIF2α antibodies. The mushroom body was identified by expression of mito-GFP. Scale bars, 20 μ m. The signal intensities of eIF2α and p-eIF2α in axons, dendrites, and cell bodies were quantified and are shown as ratios relative to the control. Means $\pm$ SE, $n = 12$. N.S., $p > 0.05$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.005$ (Student's t -test).

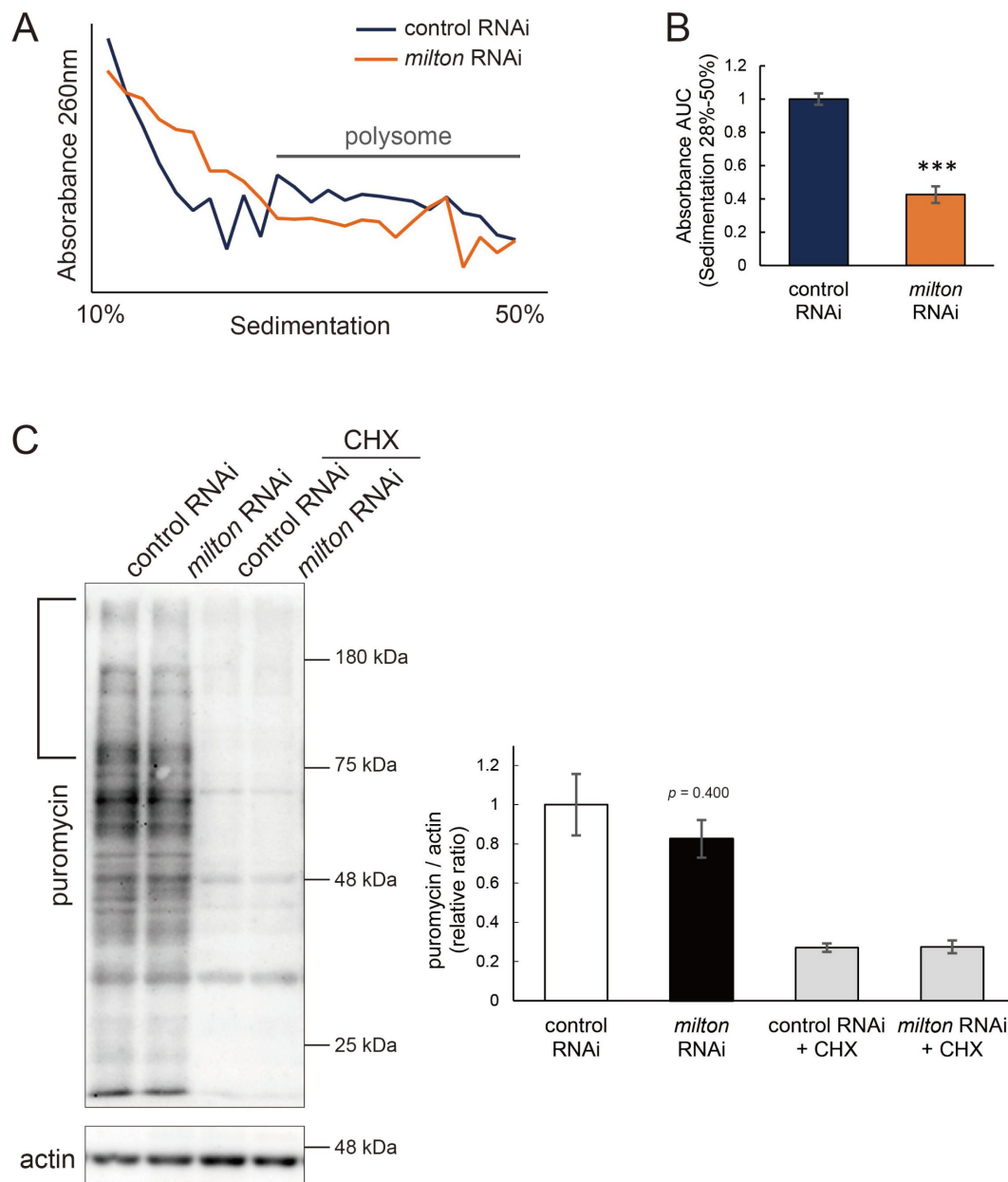


Figure 6.

***milton* knockdown suppressed global translation**

(A) Representative polysome traces of head lysates of control and *milton* knockdown flies. (B) Quantitation of polysome fraction. The relative ratio of area under the curve (AUC) of polysome fractions (sedimentation 28%-50%). Means $\pm$ SE, $n = 3$. *** $p < 0.005$ (Student's *t*-test) (C) Western blotting of head lysates of control and *milton* knockdown flies fed puromycin alone or puromycin and cycloheximide (CHX) with an anti-puromycin antibody. Flies were 14-day-old. Actin was used as a loading control. Representative blots (left) and quantitation (right) are shown. Means $\pm$ SE, $n = 3$. Student's *t*-test.

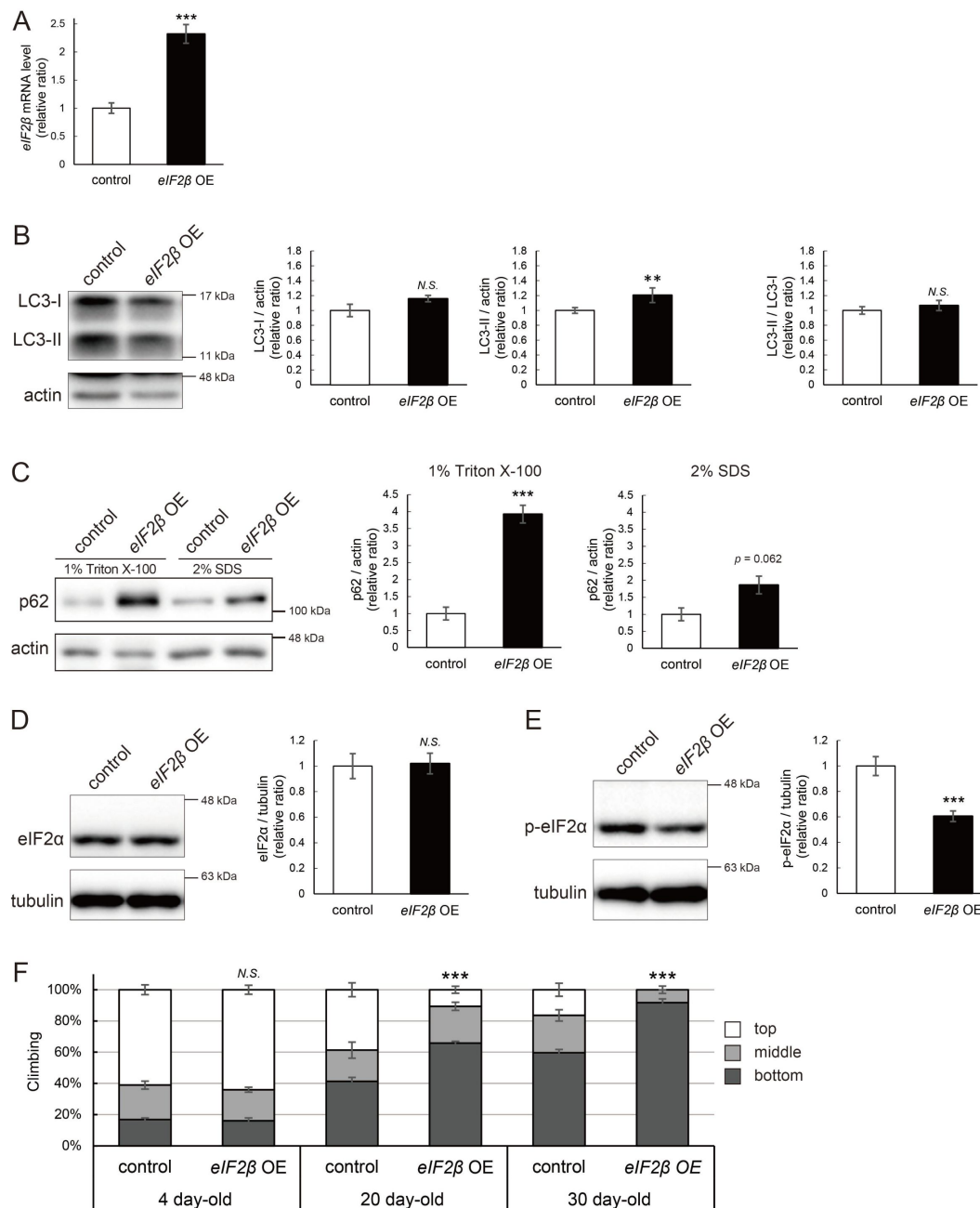


Figure 7.

eIF2β upregulation impairs autophagy and decreases locomotor function

(A) *eIF2β* mRNA levels in head extracts of flies with UAS-*eIF2β* driven by elav-Gal4 (*eIF2β* OE) or UAS-GFP driven by elav-Gal4 (control) were quantified by qRT-PCR. Flies were 2-day-old. Means $\pm$ SE, n=4. (B, C) Western blotting of head extracts with anti-LC3 (B) and anti-p62 (C) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin and actin were used as loading controls. Means $\pm$ SE, n = 3 (p62), n = 5 (LC3). (D, E) Western blotting of head extracts with anti-eIF2α (D) and anti-p-eIF2α (E) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, n = 6. (F) Climbing assay revealed early-onset of age-dependent locomotor defects in *eIF2β*-overexpressing flies. Means $\pm$ SE, n = 5. N.S., $p > 0.05$; *** $p < 0.005$ (Student's *t*-test).

during aging (**Figure 7F** [↗](#), compare 4-, 20-, and 30-day-old). We asked if *eIF2β* overexpression causes neurodegeneration, as depletion of axonal mitochondria in the photoreceptor neurons causes axon degeneration in an age-dependent manner²⁴ [↗](#). *eIF2β* overexpression in photoreceptor neurons tends to increase neurodegeneration in aged flies, while it was not statistically significant ($p > 0.05$, Figure S2).

These data indicate that an increase of *eIF2β* in neurons phenocopies depletion of axonal mitochondria, including suppression of autophagy and age-dependent locomotor dysfunction, and suggest that increase of *eIF2β* mediates these phenotypes downstream of loss of axonal mitochondria.

Lowering *eIF2β* rescues autophagic impairment and locomotor dysfunction induced by *milton* knockdown

Finally, we investigated whether suppression of *eIF2β* rescues autophagy impairment and locomotor dysfunction caused by neuronal knockdown of *milton*. Null mutants and flies with RNAi-mediated knockdown of *eIF2β* in neurons did not survive. Flies lacking one copy of the *eIF2β* gene survived without any gross abnormality, and the level of *eIF2β* mRNA in these flies was about 80% of that in control flies (**Figure 8A** [↗](#)). *eIF2β* heterozygosity did not affect the *eIF2α* and p-*eIF2α* levels (Figure S3A and B).

Neuronal knockdown of *milton* causes accumulation of autophagic substrate p62 in the Triton X-100-soluble fraction (**Figure 2B** [↗](#)), and we tested if lowering *eIF2β* ameliorates it. We found that *eIF2β* heterozygosity caused a mild increase in LC3-I levels and decreases in LC3-II levels, resulting in a significantly lower LC3-II/LC3-I ratio in *milton* knockdown flies (**Figure 8B** [↗](#)). *eIF2β* heterozygosity decreased the p62 level in the Triton X-100-soluble fraction in the brains of *milton* knockdown flies (**Figure 8C** [↗](#)). The p62 level in the SDS-soluble fraction, which is not sensitive to *milton* knockdown (**Figure 2B** [↗](#)), was not affected (**Figure 8C** [↗](#)). These results suggest that suppression of *eIF2β* ameliorates the impairment of autophagy caused by *milton* knockdown.

eIF2β heterozygosity also rescued locomotor dysfunction induced by *milton* knockdown. *milton* knockdown flies with *eIF2β* heterozygosity exhibited better locomotor function than *milton* knockdown alone (**Figure 8D** [↗](#)). The *milton* mRNA level was not increased in these flies, indicating that the rescue effect in the *eIF2β* heterozygous background was not mediated by an increase in the *milton* mRNA level (Figure S3C). These data suggest that *eIF2β* upregulation mediates autophagy impairment and locomotor dysfunction caused by the depletion of axonal mitochondria.

Discussion

The depletion of axonal mitochondria and accumulation of abnormal proteins are both characteristics of aged brains^{37,38} [↗](#). Proteostasis perturbations trigger the formation of pathological aggregates and increase the risks of neurodegenerative diseases during aging. By using neuronal *milton* knockdown to deplete mitochondria from the axon, we provide evidence that loss of axonal mitochondria drives age-related proteostasis collapse via *eIF2β* (**Figure 9** [↗](#)). We observed declines in autophagy-mediated degradation of less-aggregated proteins and proteasome activity in *milton* knockdown flies (**Figure 2** [↗](#)). Accumulation of ubiquitinated proteins and changes in age-related pathways started prematurely in *milton* knockdown flies (**Figure 1** [↗](#) and **Table 2** [↗](#)). *milton* knockdown increased *eIF2β* and lowered *eIF2α* phosphorylation in young fly brain (**Figure 4** [↗](#) and **5** [↗](#)). Overexpression of *eIF2β* phenocopied the effects of *milton* knockdown, including reduced autophagy and accelerated age-related locomotor defects (**Figure 7** [↗](#)). Furthermore, lowering *eIF2β* levels suppressed the impairment of autophagy and locomotor dysfunction induced by *milton* knockdown (**Figure 8** [↗](#)). From these

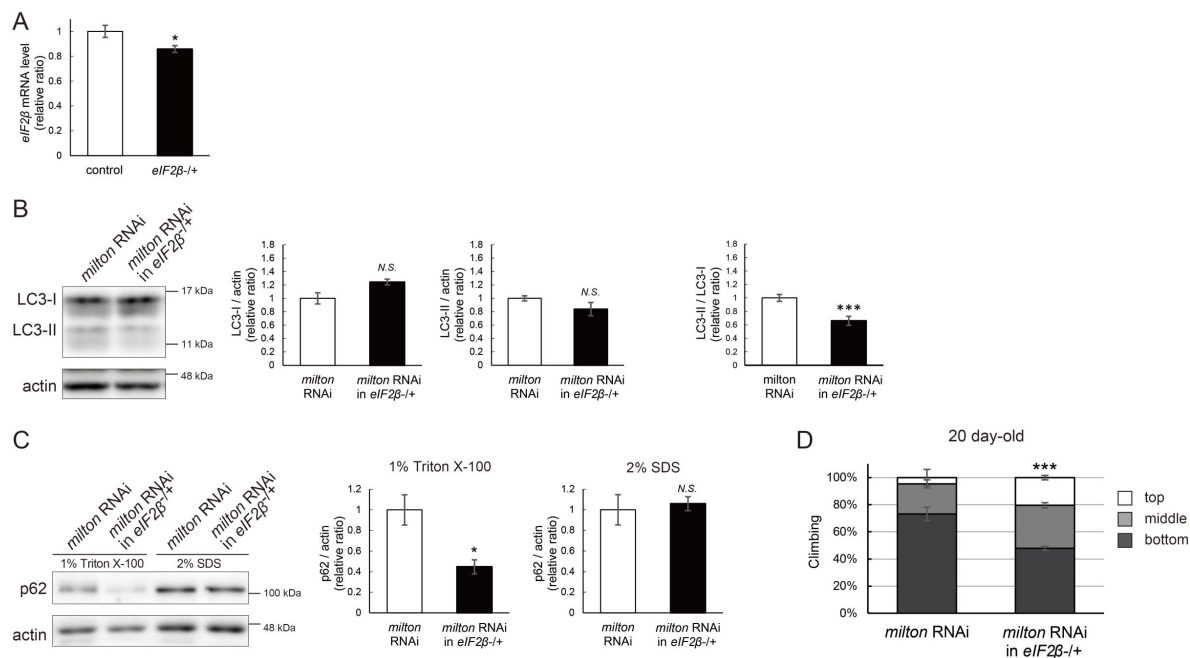


Figure 8.

Lowering *eIF2β* rescues autophagic impairment and locomotor dysfunction induced by *milton* knockdown

(A) *eIF2β* mRNA levels with one disrupted copy of the *eIF2β* gene (*eIF2β*^{SAsstopDsRed/+} (*eIF2β*^{-/+})). Head extracts of flies 2-3 day-old were analyzed by qRT-PCR. Means $\pm$ SE, $n = 3$. (B, C) Western blotting of head extracts of flies with neuronal expression of *milton* RNAi with or without *eIF2β* heterozygosity with anti-LC3 (B) and anti-p62 (C) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Actin was used as a loading control. Means $\pm$ SE, $n = 5$ (LC3), $n = 3$ (p62). (D) The climbing ability of 20-day-old flies expressing *milton* RNAi with or without *eIF2β* heterozygosity. Means $\pm$ SE, $n = 15$. N.S., $p > 0.05$; * $p < 0.05$; *** $p < 0.005$ (Student's *t*-test).

results, we propose that upregulation of eIF2 β downstream of depletion of axonal mitochondria drives age-dependent collapse of proteostasis (**Figure 9**). Our results suggest that mitochondrial distribution and eIF2 β are part of the mechanisms constituting proteostasis. *milton* knockdown causes loss of mitochondria in the axon and accumulation of mitochondria in the soma. Thus, the detrimental effects may be mediated by the accumulation of mitochondria. However, degeneration induced by *milton* knockdown is prominent in the axon and not detected in the cell body. Furthermore, abnormal protein accumulation was observed in the axon (**Figure 1**), and p-eIF2 α /eIF2 α was decreased in the neurites but not in the soma (**Figure 5**), suggesting that proteostasis defects studied in this work are caused by depletion of mitochondria rather than accumulation of mitochondria. Further analyses to dissect the effects of *milton* knockdown on proteostasis and translation in the cell body and axon by experiments with spatial resolution would be needed.

7 day-old						21 day-old					
Rank	Name	Score	Score (p) ^a	Score (v) ^b	Score (c) ^c	Rank	Name	Score	Score (p) ^a	Score (v) ^b	Score (c) ^c
1	Autophagy-related protein signaling pathway	50.394	6.76E-10	0.159	0.11	1	Histone demethylation	84.198	4.51E-28	0.102	0.422
2	Calcium signaling pathway	47.583	4.75E-15	0.146	0.117	2	CDK inhibitor signaling pathway	86.497	9.93E-18	0.078	0.292
3	Transcriptional regulation by SMAD	44.012	5.64E-14	0.146	0.095	3	Transcriptional regulation by RBE2F	46.598	9.39E-15	0.108	0.094
4	GABA signaling pathway	40.706	5.58E-13	0.122	0.123	4	Mit/Hippo signaling pathway	46.343	1.12E-14	0.09	0.133
5	Estrogen signaling pathway	37.507	5.12E-12	0.11	0.13	5	Transcriptional regulation by androgen receptor	46.078	1.35E-14	0.078	0.178
6	Sirtuin signaling pathway	36.87	7.96E-12	0.122	0.095	6	p160 SRC signaling pathway	45.809	1.62E-14	0.078	0.178
7	Transcriptional regulation by AP-1	34.874	3.18E-11	0.11	0.107	7	Transcriptional regulation by SMAD	43.961	9.84E-14	0.09	0.119
8	Arrestin signaling pathway	32.84	1.30E-10	0.11	0.092	8	Autophagy-related protein signaling pathway	41.063	4.35E-13	0.084	0.111
9	G protein (Gq/11) signaling pathway	30.889	5.03E-10	0.085	0.149	9	Transcriptional regulation by HIF	39.527	1.26E-12	0.096	0.088
10	Kainate receptor signaling pathway	30.049	9.00E-10	0.073	0.214	10	Nucleophosmin signaling pathway	38.417	2.72E-12	0.054	0.273
11	Transcriptional regulation by CREB	29.5	1.32E-09	0.098	0.093	11	HSP90 signaling pathway	37.887	3.93E-12	0.066	0.164
12	Calpain signaling pathway	28.597	2.46E-09	0.11	0.066	12	PAF metabolism	37.562	4.93E-12	0.042	0.1
13	Phospholipase D signaling pathway	28.344	2.94E-09	0.098	0.084	13	Transcriptional regulation by STAT	37.276	6.01E-12	0.072	0.130
14	HSP90 signaling pathway	27.188	6.54E-09	0.085	0.104	14	Bcl-2 family signaling pathway	36.157	1.31E-11	0.072	0.124
15	CYP family	27.188	6.54E-09	0.085	0.104	15	Sirtuin signaling pathway	34.782	3.39E-11	0.072	0.114
16	Kr3 channel signaling pathway	26.495	1.06E-08	0.061	0.25	16	Transcriptional regulation by CREB	33.819	6.60E-11	0.066	0.128
17	Estrogen biosynthesis	26.107	1.38E-08	0.061	0.238	17	PKA signaling pathway	33.172	1.03E-10	0.06	0.149
18	CaSR signaling pathway	25.39	2.27E-08	0.061	0.217	18	RSK signaling pathway	30.566	6.26E-10	0.06	0.125
19	PKA signaling pathway	24.927	3.14E-08	0.073	0.122	19	Transcriptional regulation by High mobility group protein	29.873	1.02E-09	0.054	0.148
20	PARF receptor signaling pathway	24.555	4.06E-08	0.049	0.4	20	BET family signaling pathway	29.656	1.18E-09	0.054	0.145
21	Transcriptional regulation by PPARα	24.398	4.52E-08	0.073	0.115	21	Transcriptional regulation by Myc	28.838	2.08E-09	0.066	0.093
22	BTX signaling pathway	24.398	4.52E-08	0.073	0.115	22	Transcriptional regulation by FOXO	28.827	2.10E-09	0.054	0.136
23	Transcriptional regulation by STAT	24.076	5.66E-08	0.085	0.077	23	PSD-95 family signaling pathway	26.154	1.34E-08	0.048	0.14
24	G protein (G12) signaling pathway	24.063	5.71E-08	0.073	0.111	24	AKT signaling pathway	25.169	2.65E-08	0.048	0.129
25	PARF signaling pathway	23.742	7.13E-08	0.073	0.107	25	Arginine methylation	24.799	3.43E-08	0.048	0.125
26	mGluR signaling pathway	23.742	7.13E-08	0.073	0.107	26	ap130 signaling pathway	24.25	5.01E-08	0.054	0.096
27	Free fatty acid signaling pathway	23.433	8.83E-08	0.073	0.103	27	Transcriptional regulation by CREB	23.858	6.58E-08	0.066	0.067
28	Kir channel signaling pathway	23.338	9.43E-08	0.061	0.167	28	Gene regulation by microRNAs (metastasis)	23.852	6.60E-08	0.054	0.093
29	Oxytocin signaling pathway	23.327	9.51E-08	0.049	0.333	29	HDAC signaling pathway	23.538	8.22E-08	0.036	0.207
30	Transcriptional regulation by MEF2	22.99	1.20E-07	0.073	0.098	30	Calpain signaling pathway	23.092	1.12E-07	0.06	0.074
31	S100 family signaling pathway	22.434	1.77E-07	0.073	0.092	31	Transcriptional regulation by IRF	22.738	1.43E-07	0.054	0.083
32	Transcriptional regulation by FOXO	22.301	1.94E-07	0.073	0.091	32	Disruptin signaling pathway	22.873	1.59E-07	0.048	0.104
33	P2Y signaling pathway	22.172	2.12E-07	0.061	0.143	33	14-3-3 signaling pathway	22.873	1.50E-07	0.048	0.104
34	Transcriptional regulation by SRF	21.174	4.23E-07	0.061	0.125	34	Transcriptional regulation by POU domain factor	22.601	1.57E-07	0.06	0.071
35	ATF4/ATF6/IRE1 signaling pathway	21.174	4.23E-07	0.061	0.125	35	Transcriptional regulation by BLIMP-1	22.474	1.72E-07	0.042	0.132
36	Chemokine signaling pathway	21.082	4.50E-07	0.048	0.235	36	Gene regulation by microRNAs (metabolism)	22.39	1.82E-07	0.054	0.083
37	Vasopressin signaling pathway	21.082	4.50E-07	0.048	0.235	37	Fatty acid beta oxidation	22.096	2.23E-07	0.042	0.127
38	Serotonin signaling pathway	20.854	5.28E-07	0.073	0.077	38	Transcriptional regulation by RXR	22.08	2.26E-07	0.036	0.178
39	Transcriptional regulation by HIF	20.834	5.35E-07	0.098	0.043	39	ERK signaling pathway	21.96	2.45E-07	0.048	0.098
40	Leukotriene receptors signaling pathway	20.724	5.78E-07	0.048	0.222	40	PARF signaling pathway	21.913	2.53E-07	0.042	0.125
41	CART signaling pathway	20.724	5.78E-07	0.048	0.222	41	Transcriptional regulation by VDR	21.818	3.11E-07	0.054	0.078
42	MAPK signaling pathway	20.693	5.90E-07	0.085	0.055	42	Transcriptional regulation by p53	21.188	4.24E-07	0.072	0.05
43	Transcriptional regulation by RBE2F	20.543	6.55E-07	0.098	0.042	43	Acetylcholine metabolism	21.152	4.29E-07	0.024	0.444
44	NAD metabolism	20.468	6.89E-07	0.061	0.114	44	Gene regulation by microRNAs (embryonic stem cells)	21.08	4.51E-07	0.036	0.158
45	ERK signaling pathway	20.425	7.11E-07	0.073	0.073	45	mTOR signaling pathway	21.048	4.61E-07	0.042	0.115
46	Adenyl Cyclase signaling pathway	20.393	7.73E-07	0.061	0.111	46	Gene regulation by microRNAs (cancer)	21.04	4.64E-07	0.048	0.029
47	Bile acid signaling pathway	20.141	8.68E-07	0.061	0.109	47	Transcriptional regulation by Ets-1/2	20.724	5.77E-07	0.042	0.111
						48	MAPK signaling pathway	20.411	7.18E-07	0.054	0.07
						49	Gene regulation by microRNAs (cell cycle)	20.404	7.21E-07	0.036	0.148
						50	Transcriptional regulation by p73	20.259	7.97E-07	0.042	0.109

^a Score(p) indicates p-value of the pathway.
^b Score(v) indicates the ratio of "Count" to total molecules associated with the loaded list.
^c Score(c) indicates the ratio of "Count" to total molecules contained in the pathway.

Our results suggest that the loss of axonal mitochondria is an event upstream of proteostasis collapse during aging. The number of puncta of ubiquitinated proteins was higher in *milton* knockdown at 14-day-old, but there was no significant difference at 30-day-old (**Figure 1**). Proteome analyses also showed that age-related pathways, such as immune responses, are enhanced in young flies with *milton* knockdown (**Table 2**). We also found that eIF2 β protein levels increase in an age-dependent manner until 49-day-old and reduces after that (**Figure 4G**). In the brains with neuronal knockdown of *milton*, eIF2 β levels were higher at 7-day-old than those in control and lower at the 21-day-old (**Figure 4D** and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude. Disruption of proteostasis is expected to contribute neurodegeneration³⁸, and it would be interesting to analyze the sequence of protein accumulation and axonal degeneration in *milton* knockdown (²⁴,²⁹ and **Figure 1**) in detail with higher time resolution.

Our results revealed that eIF2 β regulates autophagy and maintains proteostasis during aging. eIF2 β is a component of eIF2, which mediates translational regulation and ISR initiation. When ISR is activated, phosphorylated eIF2 α suppresses global translation and induces translation of ATF4, which mediates transcription of autophagy-related genes^{39,40}. Since ISR can positively regulate autophagy, we suspected that suppression of ISR underlies a reduction in autophagic protein degradation. We found neuronal knockdown of *milton* reduced phosphorylated eIF2 α ,

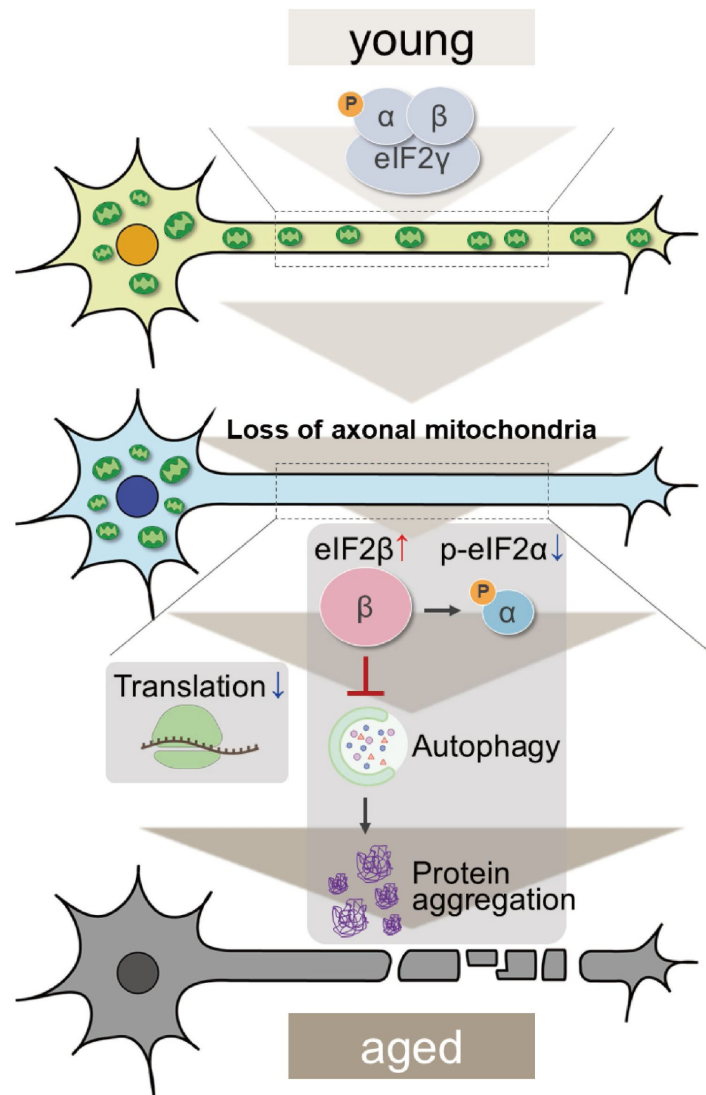


Figure 9.

Graphical abstract: the mitochondria-eIF2 β axis in the axon maintains neuronal proteostasis during aging

suggesting that ISR is reduced (**Figure 5**). However, we also found that global translation was reduced (**Figure 6**). Increased levels of eIF2 β might disrupt the eIF2 complex or alter its functions. The stoichiometric mismatch caused by an imbalance of eIF2 components may inhibit ISR induction. Supporting this model, we found that eIF2 β upregulation reduced the levels of p-eIF2 α (**Figure 7**). It is also possible that eIF2 β mediates autophagy defects via mechanisms independent of ISR since eIF2 β has functions independent of eIF2^{41,42}. For example, suppression of *eIF2 β* has been reported to slow down cancer cell growth⁴¹. In developing neurons, eIF2 β can directly interact with the translational repressor Kra to regulate midline axon guidance⁴². However, to our knowledge, the roles of eIF2 β in aging have not been reported. Our results revealed a novel function of eIF2 β to maintain proteostasis during aging.

How depletion of axonal mitochondria upregulates eIF2 β is currently under investigation. A major mitochondrial function is ATP production, and depletion of axonal mitochondria downregulates ATP in axons³³. However, we found that ATP deprivation did not always suppress autophagy (**Figure 3**), suggesting it is unlikely to be involved in the mechanisms that induce eIF2 β upregulation. Mitochondria also serve as signaling hubs for translation and protein degradation. Mitochondrial proteins are regulated by co-translational protein quality control, and mitochondrial damage induces translational stalling of mitochondrial outer membrane-associated *complex-I 30 kD subunit (C-I30)* mRNA⁴³. Additionally, the mitochondrial outer membrane ubiquitin ligase MITOL (also known as MARCHF5) ubiquitinates and regulates not only mitochondrial proteins such as Mfn2⁴⁴ but also microtubule-associated⁴⁵ and endoplasmic reticulum⁴⁶ proteins. These findings indicate that mitochondria serve as local signaling centers for proteostasis maintenance, and eIF2 β levels may also be regulated by mechanisms related to mitochondria.

In conclusion, our results suggest that axonal mitochondria and eIF2 β form an axis to maintain constitutive autophagy. Suppression of *eIF2 β* rescued autophagic defects and neuronal dysfunction upon loss of axonal mitochondria. Since eIF2 β is conserved across many species, including *Drosophila* and humans, our results suggest that eIF2 β may be a possible therapeutic target for aging and diseases associated with mitochondrial mislocalization.

Materials and methods

Fly stocks and husbandry

Flies were maintained in standard cornmeal medium (10% glucose, 0.7% agar, 9% cornmeal, 4% yeast extract, 0.3% propionic acid, and 0.1% n-butyl p-hydroxybenzoate) at 25°C under light–dark cycles of 12:12 h. The flies were transferred to fresh food vials for every 2–3 days. UAS-*milton* RNAi (v41508) was from VDRC and outcrossed to [w1118] for five generations in our laboratory. Transgenic fly lines carrying UAS-*luciferase* RNAi control for *milton* RNAi was reported previously²⁴, GMR-gal4, Elav-gal4, UAS-*Pfk* RNAi (Bloomington stock center #36782), UAS-*luciferase* RNAi (Bloomington stock center #31603) (control for *Pfk* RNAi), UAS-*GFP* (used for control for UAS-*eIF2 β*) and UAS-*eIF2 β* (eIF2 β ^{EY08063}, Bloomington stock center #17425) were from the Bloomington stock center. *eIF2 β* heterozygous strain (PBac{*SastopDsRed*} LL07719, DGRC#142114) was from KYOTO *Drosophila* Stock Center. UAS-*mitoGFP* was a kind gift from Drs. W. M. Saxton (University of California, Santa Cruz).

Immunohistochemistry and image acquisition

Fly brains were dissected in PBS and fixed for 45 minutes in formaldehyde (4% v/v in PBS) at room temperature. After incubation in PBST containing 0.1% Triton X-100 for 10 min three times, samples were incubated for 1h at room temperature in PBST containing 1% normal goat serum (Wako, #143-06561) and then incubated overnight with the primary antibody (anti-ubiquitin antibody Ubi-1 (Thermo Fisher #13-1600) (1:50), anti-eIF2 α (abcam #ab26197) (1:50) and anti-p-

eIF2 α (Cell signaling #3398S) (1:50) diluted in 1% NGS/PBST at 4 °C. Samples were then washed for 10 min with PBST including 0.1% Triton X-100 three times and incubated with the secondary antibody for overnight at 4° C. Brains were mounted in Vectashield (Vectorlab Cat#H-1100) and analyzed under a confocal microscope (Nikon). Quantitative analysis was performed using ImageJ (National Institutes of Health) with maximum projection images derived from Z-stack images acquired with same settings. Puncta were identified with mean intensity and area using ImageJ. For eIF2 α and p- eIF2 α immunostaining, the mushroom body was detected by mitoGFP expression.

Electron microscopy

Proboscis was removed from decapitated heads, which were then incubated in primary fixative solution (2.5% glutaraldehyde and 2% paraformaldehyde in 0.1 M sodium cacodylate buffer) at R.T. for 2 hours. After washing heads with 3% sucrose in 0.1 M sodium cacodylate buffer, fly heads were post-fixed for 1 hour in secondary fixation (1% osmium tetroxide in 0.1 M sodium cacodylate buffer) on ice. After washing with H₂O, heads were dehydrated with ethanol and infiltrated with propylene oxide and Epon mixture (TAAB and Nissin EM) for 3 hours. After infiltration, specimens were embedded with an Epon mixture at 70°C for 2-3 days. Thin sections (70 nm) of laminas were collected on copper grids. The sections were stained with 5% uranyl acetate in 50% ethanol and Reynolds' lead citrate solution. Electron micrographs were obtained with a CCD Camera mounted on a JEM-1400 plus electron microscope (Jeol Ltd.). Quantitation was performed using ImageJ (National Institutes of Health).

SDS-PAGE and immunoblotting

Western blotting was performed as reported previously²⁴. Briefly, heads of 10-20 *Drosophila* were homogenized with SDS-Tris-Glycine sample buffer (0.312M Tris, 5% SDS, 8% glycerol, 0.0625% BPB, 10% β -mercaptoethanol, 10 μ g/mL leupeptin, 0.4 μ M Pefabloc, 10mM β -glycerphosphate, 10mM NaF) and after boiling at 95°C for 2 minutes, it was centrifuged at 13,200 rpm, and the supernatant was used as a sample. For p62 western blot, fly heads were homogenized with 1% PBST and after centrifugation at 13,200 rpm, the supernatant was mixed 1:1 SDS-Tris-Glycine sample buffer, and boiled at 95 °C for 2 minutes. The pellet was dissolved with 2% SDS in PBS, then centrifuged again at 13,200 rpm. The supernatant was mixed 1:1 SDS-Tris-Glycine sample buffer and then boiled at 95 °C for 2 minutes. SDS-PAGE for western blotting was performed using 15%(w/v) (LC3), 7.5%(w/v) (p62), 10% (w/v) (eIF2 α , β , and p-eIF2 α) polyacrylamide gels. After electrophoresis, they were transferred to PVDF membrane (Merck Millipore) using a transfer device (BIO-RAD). After transfer, the membrane was blocked with 5% skim milk/TBST (50 mM Tris (pH 7.5), 0.15 M NaCl, 0.05% Tween20) for 1 hour and incubated with primary antibody listed below overnight at 4 °C. Membranes were rinsed twice with TBST containing 0.65M NaCl and once with TBST containing 0.15M NaCl. After incubation with the secondary antibody at room temperature for 1 hour, membranes were rinsed twice with TBST containing 0.65M NaCl and once with TBST containing 0.15M NaCl. After incubation with Immobilon Western Chemiluminescent HRP Substrate (Merck Millipore), chemiluminescent signals were detected with Fusion FX (Vilber). Experiments were repeated at least 3 times with independent cohorts of flies.

Primary antibodies: anti-LC3 antibody Atg8 (Merck Millipore #ABC975) (1:1000), anti-p62 antibody Ref2P (abcam #ab178440) (1:750), anti-eIF2 β antibody (1:1500), anti-eIF2 α antibody (abcam #ab26197) (1:1000), anti-p-eIF2 α antibody (Cell signaling #3398S) (1:2000), anti-actin antibody (SIGMA A2066) (1:3000) and anti- β tubulin antibody (Sigma #T9026) (1:800,000). Polyclonal anti-eIF2 β antibody was raised against a synthetic peptide (CGLEDDTKKEDPQDEA) corresponding to the C-terminal residues 29-43 of *Drosophila* eIF2 β (1:1500).

Secondary antibodies: Peroxidase-conjugated goat anti-mouse IgG antibody (Dako #P0447) (1:2000), peroxidase-conjugated pig anti-rabbit IgG antibody (Dako #P0399) (1:2000)

Proteasome assay

Heads from ten flies were homogenized in 150 μ l of buffer B (25 mM Tris-HCl [pH 7.5], 2 mM ATP, 5 mM MgCl₂, and 1 mM dithiothreitol). Proteasome peptidase activity in the lysates was measured with a synthetic peptide substrate, succinyl-Leu-Leu-Val-Tyr-7-amino-4-methyl-coumarin (Suc-LLVY-AMC) (Cayman). Luminescence was measured on a multimode plate reader 2300 Enspire (PerkinElmer). Experiments were repeated at least 3 times with independent cohorts of flies.

ATP assay

Heads from the 10 flies were homogenized in 50 μ l of 6 M guanidine-HCl in extraction buffer (100 mM Tris and 4 mM EDTA, pH 7.8) to inhibit ATPases. Samples were boiled for 5 min and centrifuged. The supernatant was diluted 4% with extraction buffer and mixed with a reaction solution (ATP Determination kit, Invitrogen). Luminescence was measured on a multimode plate reader 2300 Enspire (PerkinElmer). The relative ATP levels were calculated by dividing the luminescence by the total protein concentration, which was determined by the Bradford method. Experiments were repeated at least 3 times with independent cohorts of flies.

Proteomic assay and pathway analysis

Sample preparation

Heads from the 35 flies were homogenized in 110 μ l of extraction buffer (0.25% RapiGest SF, 50mM ammonium bicarbonate, 10mM dithiothreitol, 10 μ g/mL leupeptin, 0.4 μ M Pefabloc, 10mM β -glycerphosphate, 10mM NaF). Homogenized samples were centrifuged and boiled for 5 min. After quantification of the protein concentration using a Pierce® 660 nm Protein Assay (Thermo Fisher Scientific), 10 μ g proteins from each sample were reduced using 5 mM tris (2-carboxyethyl) phosphine hydrochloride (TCEP-HCl; Thermo Fisher Scientific) at 60°C for 1 h, alkylated using 15 mM iodoacetamide (Fujifilm Wako Pure Chemical, Osaka, Japan) at room temperature for 30 min, and then digested using 1.5 μ g Trypsin Gold (Mass Spectrometry Grade; Promega, Madison, WI, USA) at 37°C for 17 h. The digests were acidified by the addition of trifluoroacetic acid (TFA), incubated at 37°C for 30 min, and then centrifuged at 17,000 $\times$ g for 10 min to remove the RapiGest SF. The supernatants were collected and desalted using MonoSpin™ C18 (GL Sciences, Tokyo, Japan). The resulting eluates were concentrated *in vacuo*, dissolved in 2% MeCN containing 0.1% formic acid (FA), and subjected to LC-MS/MS analysis.

LC-MS/MS analysis and database search

LC-MS/MS analyses were performed on an Ultimate 3000 RSLCnano system (Thermo Fisher Scientific) coupled to a Q Exactive hybrid quadrupole-Orbitrap mass spectrometer (Thermo Fisher Scientific) equipped with a nano electron spray ionization (ESI) source. The LC system was equipped with a trap column (C18 PepMap 100, 0.3 $\times$ 5 mm, 5 μ m, Thermo Fisher Scientific) and an analytical column (NTCC-360/75-3-125, Nikkyo Technos, Tokyo, Japan). Peptide separation was performed using a 90-min gradient of water/0.1% FA (mobile phase A) and MeCN/0.1% FA (mobile phase B) at a flow rate of 300 nL/min. Elution was performed as follows: 0–3 min, 2% B; 3–93 min, 2%–40% B; 93–95 min, 40%–95% B; 95–105 min, 95% B; 105–107 min, 95%–2% B; and 107–120 min, 2% B. The mass spectrometer was operated in data-dependent acquisition mode. The MS parameters were as follows: spray voltage, 2.0 kV; capillary temperature, 275°C; S-lens RF level, 50; scan type, full MS; scan range, m/z 350–1500; resolution, 70,000; polarity, positive; automatic gain control target, 3×10^6 ; and maximum injection time, 100 msec. The MS/MS parameters were as follows: resolution, 17,500; automatic gain control target, 1×10^5 ; maximum injection time, 60 msec; normalized collision energy (NCE), 27; dynamic exclusion, 15 sec; loop count, 10; isolation window, 1.6 m/z ; charge exclusion: unassigned, 1 and ≥ 8 ; and injection volume, 1 μ L (containing 0.5 μ g protein). Measurements were made in duplicate for each sample.

The identification of proteins and label-free quantification (LFQ) of the detected peptides were performed using Proteome Discoverer software ver. 2.4 (Thermo Fisher Scientific). The analytical parameters used for the database search were as follows: parent mass error tolerance, 10.0 ppm; fragment mass error tolerance, 0.02 Da; search engine, sequest HT; protein database, *Drosophila melanogaster* (Fruit fly: SwissProt Tax ID=7227); enzyme name, trypsin (full); maximum number of missed cleavages, 2; dynamic modification, oxidation (methionine), phosphorylation (serine, threonine, tyrosine), acetyl (lysine), GG (lysine); N-terminal modification, Met-loss (methionine), and Met-loss+acetyl (methionine); static modification, carbamidomethylation (cysteine) and FDR confidence, High < 0.01, 0.01 ≤ Medium < 0.05, 0.05 ≤ Low. The parameters for LFQ were as follows: precursor abundance, based on area; and normalization mode, total peptide amount.

The abundance ratio of *milton* RNAi to control RNAi at 7- or 21-day-old was calculated. We considered proteins with an abundance ratio of ≥2.0 or ≤ 0.5 and an ANOVA *P*-value of < 0.05 based on volcano plots to be differentially expressed of *milton* RNAi. To extract molecular networks biologically relevant to the proteins that are differentially expressed in *milton* RNAi, pathway analysis was performed using KeyMolnet (KM Data Inc., Tokyo, Japan).

RNA extraction and quantitative real-time PCR analysis

Heads from more than 25 flies were mechanically isolated, and total RNA was extracted using ISOGEN (NipponGene) followed by reverse-transcription using PrimeScript RT reagent kit (Takara). The resulting cDNA was used as a template for PCR with THUNDERBIRD SYBR qPCR mix (TOYOBO) on a Thermal Cycler Dice real-time system TP800 (Takara). Expression of genes of interest was standardized relative to *rp49*. Relative expression values were determined by the $\Delta\Delta CT$ method (Livak and Schmittgen, 2001). Experiments were repeated three times, and a representative result was shown.

Primers were designed using DRSC FlyPrimerBank (Harvard Medical School). Primer sequences are shown below;

eIF2β for 5'-GGACGACGACAAGAGCGAAG -3'

eIF2β rev 5'-CGGTCGCATCACGAACCTTG -3'

milton for 5'-GGCTTCAGGGCCAGGTATCT-3'

milton rev 5'-GCCGAACTTGGCTGACTTTG-3'

Actin for 5'-TGCACCGCAAGTGCTTCTAA-3'

Actin rev 5'-TGCTGCACTCCAACTTCCA-3'

rp49 for 5'-GCTAAGCTGTCGCACAAATG-3' *rp49* rev 5'- GTTCGATCCGTAACCGATGT-3'

Polysome gradient centrifugation

30 heads were homogenized in 150 μ l of lysis buffer (25 mM Tris pH 7.5, 50 mM MgCl₂, 250 mM NaCl, 1 mM DTT, 0.5 mg/ml cycloheximide, 0.1 mg/ml heparin). The lysates were centrifuged at 13,200 rpm at 4 °C for 5 minutes, and the supernatant was collected. The samples containing 38 μ g of RNA was layered gently on top of a 10–50% w/w sucrose gradient (50 mM Tris pH 7.5, 50 mM MgCl₂, 250 mM NaCl, 0.1 mg/ml heparin, 0.5 mg/ml cycloheximide in 5 ml polyallomer tube) and centrifuged at 37,000 rpm at 4°C for 150 minutes in a himac CP-NX ultracentrifuge using a P50AT rotor. Samples were fractionated from top to bottom, and absorbance at OD260 nm was analyzed by a Plate reader (EnSpire). Experiments were repeated at least 3 times with independent cohorts of flies.

Puromycin analysis

13-day-old flies were starved for 6 hr and fed 600 μ M puromycin (Sigma) or 600 μ M puromycin/35 mM cycloheximide (Sigma) in 5% sucrose solution for 20 hr. Incorporated puromycin was quantified by western blot with anti-puromycin antibody and normalized with actin. Experiments were repeated at least 3 times with independent cohorts of flies.

Histological analysis

Fly heads were fixed in Bouin's fixative solution for 48 h at room temperature, incubated for 24 h in 50 mM Tris/150 mM NaCl, and embedded in paraffin. Serial sections (7 μ m thickness) through the entire heads were stained with hematoxylin and eosin and examined by bright-field microscopy. Images of the sections that include the lamina were captured with Keyence microscope BZ-X700 (Keyence), and the vacuole area was measured using ImageJ (National Institutes of Health).

Climbing assay

The climbing assay was performed as previously described²⁴[\[14\]](#). Flies were placed in an empty plastic vial (2.5 cm in diameter $\times$ 10 cm in length). The vial was gently tapped to knock the flies to the bottom, and the number of flies reached to the top, middle and bottom areas of the vials in 10 s was counted. Experiments were repeated 10 times, and the mean percentage of flies in each area and standard deviations were calculated. Experiments were repeated with independent cohorts more than three times, and a representative result was shown.

Statistics

The number of replicates, what n represents, precision measurements, and the meaning of error bars are indicated in Figure Legends. Data are shown as means $\pm$ SEM. For pairwise comparisons, Student's t-test was performed with Microsoft Excel (Microsoft). For multiple comparisons, data were analyzed using one-way ANOVA with Tukey's HSD multiple-comparisons test in the GraphPad Prism 6.0 software (GraphPad Software, Inc., La Jolla, CA). Results with a p-value of less than 0.05 were considered to be statistically significant.

Data Availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon request.

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Additional information

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Author contributions

K.S. and K.A. designed the experiments, interpreted the data, and wrote the manuscript; K.S., Y.M., K.M.I and E.S. conducted the experiments, and K.S., Y.M., E.S., and K.A. analyzed the data. All authors read and approved the manuscript.

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Additional files

Figure S1. Supplemental Figure 1. Ubiquitinated proteins in brains with neuronal knockdown of milton at 1-day-old. Brains dissected at 1-day-old were immunostained with an antibody against ubiquitin. Representative images (left) and quantitation of the number of ubiquitin-positive puncta (right) are shown. Scale bars of hemibrains, 100 μ m, Means $\pm$ SE, $n = 8$. N.S., $p > 0.05$ (Student's t -test). [↗](#)

Figure S2. Supplemental Figure 2. Histology analysis of fly heads with eIF2 β overexpression. The morphology of the eye with eIF2 β overexpression. The dotted lines indicate the retina. Vacuoles are indicated by arrows. Representative images (left) and quantification of vacuole area (right). The flies were 40-day-old. Means $\pm$ SE, $n = 5-13$ N.S., $p > 0.05$ (Student's t -test). [↗](#)

Figure S3. Supplemental Figure 3. Lowering the eIF2 β level does not affect the levels of eIF2 α and p-eIF2 α . (A, B) Blotting was performed with anti-eIF2 α (A) and anti-p-eIF2 α (B) antibodies. Flies were 14-day-old. Representative blots (left) and quantitation (right) are shown. Tubulin was used as a loading control. Means $\pm$ SE, $n = 6$. (C) eIF2 β gene disruption does not affect the knockdown efficiency of milton. milton mRNA levels in head extracts were quantified by qRT-PCR. Flies were 2-day-old. Means $\pm$ SE, $n = 3$. N.S., $p > 0.05$; * $p < 0.05$ (Student's t -test). [↗](#)

Supplementary tables 1 and 2. [↗](#)

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Reviewer #1 (Public review):

The study presents significant findings on the role of mitochondrial depletion in axons and its impact on neuronal proteostasis. It effectively demonstrates how the loss of axonal mitochondria and elevated levels of eIF2 β contribute to autophagy collapse and neuronal dysfunction. The use of *Drosophila* as a model organism and comprehensive proteome analysis adds robustness to the findings.

In this revision, the authors have responded thoughtfully to previous concerns. In particular, they have addressed the need for a quantitative analysis of age-dependent changes in eIF2 β and eIF2 α . By adding western blot data from multiple time points (7 to 63 days), they show that eIF2 β levels gradually increase until middle age, then decline. In milton knockdown flies, this pattern appears shifted, supporting the idea that mitochondrial defects may accelerate aging-related molecular changes. These additions clarify the temporal dynamics of eIF2 β and improve the overall interpretation.

Other updates include appropriate corrections to figures and quantification methods. The authors have also revised some of their earlier mechanistic claims, presenting a more cautious interpretation of their findings.

Overall, this work provides new insights into how mitochondrial transport defects may influence aging-related proteostasis through eIF2 β . The manuscript is now more convincing, and the revisions address the main points raised earlier. I find the updated version much improved.

<https://doi.org/10.7554/eLife.95576.3.sa2>

Reviewer #2 (Public review):

In the manuscript, the authors aimed to elucidate the molecular mechanism that explains neurodegeneration caused by the depletion of axonal mitochondria. In *Drosophila*, starting with siRNA depletion of Milton and Miro, the authors attempted to demonstrate that the depletion of axonal mitochondria induces the defect in autophagy. From proteome analyses, the authors hypothesized that autophagy is impacted by the abundance of eIF2 β and the phosphorylation of eIF2 α . The authors followed up the proteome analyses by testing the effects of eIF2 β overexpression and depletion on autophagy. With the results from those experiments, the authors proposed a novel role of eIF2 β in proteostasis that underlies neurodegeneration derived from the depletion of axonal mitochondria, which they suggest accelerates age-dependent changes rather than increasing their magnitude.

Strong caution is necessary regarding the interpretation of translational regulation resulting from the Milton KD. The effect of Milton KD on translation appears subtle, if present at all, in the puromycin incorporation experiments in both the initial and revised versions. Additionally, the polysome profiling data in the revised manuscript lack the clear resolution for ribosomal subunits, monosomes, and polysomes that is typically expected in publications.

<https://doi.org/10.7554/eLife.95576.3.sa1>

Author response:

The following is the authors' response to the previous reviews.

Reviewer #1 (Public Review):

The authors observed a decline in autophagy and proteasome activity in the context of Milton knockdown. Through proteomic analysis, they identified an increase in the protein levels of eIF2 β , subsequently pinpointing a novel interaction within eIF subunits where eIF2 β contributes to the reduction of eIF2 α phosphorylation levels. Furthermore, they demonstrated that overexpression of eIF2 β suppresses autophagy and leads to diminished motor function. It was also shown that in a heterozygous mutant background of eIF2 β , Milton knockdown could be rescued. This work represents a novel and significant contribution to the field, revealing for the first time that the loss of mitochondria from axons can lead to impaired autophagy function via eIF2 β , potentially influencing the acceleration of aging.

Thank you so much for your review and comments.

Reviewer #2 (Public Review):

*In the manuscript, the authors aimed to elucidate the molecular mechanism that explains neurodegeneration caused by the depletion of axonal mitochondria. In *Drosophila*, starting with siRNA depletion of Milton and Miro, the authors attempted to demonstrate that the depletion of axonal mitochondria induces the defect in autophagy. From proteome analyses, the authors hypothesized that autophagy is impacted by the abundance of eIF2 β and the phosphorylation of eIF2 α . The authors followed up the proteome analyses by testing the effects of eIF2 β overexpression and depletion on autophagy. With the results from those experiments, the authors proposed a novel role of eIF2 β in proteostasis that underlies neurodegeneration derived from the depletion of axonal mitochondria.*

The manuscript has several weaknesses. The reader should take extra care while reading this manuscript and when acknowledging the findings and the model in this manuscript.

The defect in autophagy by the depletion of axonal mitochondria is one of the main claims in the paper. The authors should work more on describing their results of LC3-II/LC3-I ratio, as there are multiple ways to interpret the LC3 blotting for the autophagy assessment. Lysosomal defects result in the accumulation of LC3-II thus the LC3-II/LC3-I ratio gets higher. On the other hand, the defect in the early steps of autophagosome formation could result in a lower LC3-II/LC3-I ratio. From the results of the actual blotting, the LC3-I abundance is the source of the major difference for all conditions (Milton RNAi and eIF2 β overexpression and depletion).

Thank you so much for your review and comments. As the reviewer pointed out, LC3-II/LC3-I ratio changes do not necessarily indicate autophagy defects. However, since p62 accumulation (Figure 2B, 2E, 3E, Figure 8C, Figure 9C), these results collectively suggest that autophagy is lowered.

As the reviewer pointed out and we described in v2, milton knockdown, eIF2 β overexpression and heterozygosity increase LC3-I abundance. We do not know how these conditions increase LC3-I at this moment. We will investigate the cause of the increase in LC3-I by milton knockdown and how it contribute to impaired autophagy. We added this discussion as:

Lines 388-393: 'Our results also suggest that milton knockdown and overexpression of eIF2 β affect autophagy via increased LC3-I abundance (Figures 2 and 7), suggesting an unconventional mechanism of autophagy suppression. To our knowledge, the roles of eIF2 β in aging and autophagy independent of ISR have not been reported. Our results revealed a novel function of eIF2 β to maintain proteostasis during aging, while further investigation is required to elucidate underlying mechanisms.'

Another main point of the paper is the up-regulation of eIF2 β by depleting the axonal mitochondria leads to the proteostasis crisis. This claim is formed by the findings from the proteome analyses. The authors should have presented their proteomic data with much thorough presentation and explanation. As in the experiment scheme shown in Figure 4A, the author did two proteome analyses: one from the 7-day-old sample and the other from the 21-day-old sample. The manuscript only shows a plot of the result from the 7-day-old sample, but that of the result from the 21-day-old sample. For the 21-day-old sample, the authors only provided data in the supplemental table, in which the abundance ratio of eIF2 β from the 21-day-old sample is 0.753, meaning eIF2 β is depleted in the 21-day-old sample. The authors should have explained the impact of the eIF2 β depletion in the 21-day-old sample, so the reader could fully understand the authors' interpretation of the role of eIF2 β on proteostasis.

Thank you for pointing it out. Plots of the 21-day-old proteome results was included in the main figure (Figure 4C) in v2. In this revision, we further analyzed age-dependent changes of eIF2 β levels by western blotting (Figure 4G). We found that eIF2 β levels increased during aging until 49-day-old then reduced at 63-day-old (Figure 4G in the revised manuscript). At the young age, eIF2 β levels were higher in milton knockdown brain compared to the control, and eIF2 β levels were lower in milton knockdown brains than those in the control. These results suggest that *milton* knockdown accelerates age-dependent changes in eIF2 β . We added these results and discussion in the revised manuscript.

Lines 240-243: 'We also investigated age-dependent changes in eIF2 β by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. eIF2 β levels increased during aging until

49-day-old (Figure 4G). These results suggest that upregulation of eIF2 β in *milton* knockdown fly brain reflects early an onset of age-dependent increase of eIF2 β levels.’

Lines 363-368: ‘We also found that eIF2 β protein levels increase in an age-dependent manner until 49-day-old and reduces after that (Figure 4G). In the brains with neuronal knockdown of *milton*, eIF2 β levels were higher at 7-day-old than those in control and lower at the 21-day-old (Figure 4D and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude.’ Our new data indicate that eIF2 β levels increase during aging in control flies until 49-day-old, then reduce at 63-day-old (included as Figure 4G in the revised manuscript). These age-dependent changes might explain the reduction in eIF2 β levels in *Milton* knockdown compared to the control in middle age: higher eIF2 β levels in *milton* knockdown flies at a young age than control and lower eIF2 β levels in the middle-aged flies may reflect premature aging.

We included these sentences in the discussion section:

Lines 240-243: ‘We also investigated age-dependent changes in eIF2 β by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. eIF2 β levels increased during aging until 49-day-old (Figure 4G). These results suggest that upregulation of eIF2 β in *milton* knockdown fly brain reflects early an onset of age-dependent increase of eIF2 β levels.’

Lines 359-371: ‘Our results suggest that the loss of axonal mitochondria is an event upstream of proteostasis collapse during aging. The number of puncta of ubiquitinated proteins was higher in *milton* knockdown at 14-day-old, but there was no significant difference at 30-day-old (Figure 1). Proteome analyses also showed that age-related pathways, such as immune responses, are enhanced in young flies with *milton* knockdown (Table 2). We also found that eIF2 β protein levels increase in an age-dependent manner until 49-day-old and reduces after that (Figure 4G). In the brains with neuronal knockdown of *milton*, eIF2 β levels were higher at 7-day-old than those in control and lower at the 21-day-old (Figure 4D and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude. Disruption of proteostasis is expected to contribute neurodegeneration³⁸, and it would be interesting to analyze the sequence of protein accumulation and axonal degeneration in *milton* knockdown (24,29 and Figure 1) in detail with higher time resolution.’

With our new data, we revised some of our responses to the first round of reviewer’s comments.

Reviewer #1 (Public Review):

The authors observed a decline in autophagy and proteasome activity in the context of Milton knockdown. Through proteomic analysis, they identified an increase in the protein levels of eIF2 β , subsequently pinpointing a novel interaction within eIF subunits where eIF2 β contributes to the reduction of eIF2 α phosphorylation levels. Furthermore, they demonstrated that overexpression of eIF2 β suppresses autophagy and leads to diminished motor function. It was also shown that in a heterozygous mutant background of eIF2 β , Milton knockdown could be rescued. This work represents a novel and significant contribution to the field, revealing for the first time that the loss of mitochondria from axons can lead to impaired autophagy function via eIF2 β , potentially influencing the acceleration of aging. To further support the authors’ claims, several improvements are necessary, particularly in the methods of quantification and the points that should be demonstrated quantitatively. It is crucial to investigate the correlation between aging and the proteins eIF2 β and eIF2 α .

Thank you so much for your review and comments. We included analyses of protein levels of eIF2 α , eIF2 β , and eIF2 γ at 7 days and 21 days (Figure 4D). The manuscript was revised as below;

Lines 246-249 ‘As for the other subunits of eIF2 complex, proteome analysis did not detect a significant difference in the protein levels of eIF2 α and eIF2 γ between *milton* knockdown and control flies at 7 and 21 days (Figure 4D).’

NEW TEXT: We analyzed age-dependent changes of eIF2 β levels in more detail by western blotting (Figure 4G). We found that eIF2 β levels increased during aging until 49-day-old then reduced at 63-day-old (Figure 4G in the revised manuscript). At the young age, eIF2 β levels were higher in *milton* knockdown brain compared to the control, and eIF2 β levels were lower in *milton* knockdown brains than those in the control. These results suggest that *Milton* knockdown accelerates age-dependent changes in eIF2 β . We added these results and discussion in the revised manuscript.

NEW TEXT: Lines 240-243: ‘We also investigated age-dependent changes in eIF2 β by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. eIF2 β levels increased during aging until 49-day-old (Figure 4G). These results suggest that upregulation of eIF2 β in *milton* knockdown fly brain reflects early an onset of age-dependent increase of eIF2 β levels.’

NEW TEXT: Lines 363-368: ‘We also found that eIF2 β protein levels increase in an age-dependent manner until 49-day-old and reduces after that (Figure 4G). In the brains with neuronal knockdown of *milton*, eIF2 β levels were higher at 7-day-old than those in control and lower at the 21-day-old (Figure 4D and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude.’

Reviewer #2 (Public Review):

In the manuscript, the authors aimed to elucidate the molecular mechanism that explains neurodegeneration caused by the depletion of axonal mitochondria. In Drosophila, starting with siRNA depletion of Milton and Miro, the authors attempted to demonstrate that the depletion of axonal mitochondria induces the defect in autophagy. From proteome analyses, the authors hypothesized that autophagy is impacted by the abundance of eIF2 β and the phosphorylation of eIF2 α . The authors followed up the proteome analyses by testing the effects of eIF2 β overexpression and depletion on autophagy. With the results from those experiments, the authors proposed a novel role of eIF2 β in proteostasis that underlies neurodegeneration derived from the depletion of axonal mitochondria.

The manuscript has several weaknesses. The reader should take extra care while reading this manuscript and when acknowledging the findings and the model in this manuscript.

The defect in autophagy by the depletion of axonal mitochondria is one of the main claims in the paper. The authors should work more on describing their results of LC3-II/LC3-I ratio, as there are multiple ways to interpret the LC3 blotting for the autophagy assessment. Lysosomal defects result in the accumulation of LC3-II thus the LC3-II/LC3-I ratio gets higher. On the other hand, the defect in the early steps of autophagosome formation could result in a lower LC3-II/LC3-I ratio. From the results of the actual blotting, the LC3-I abundance is the source of the major difference for all conditions (Milton RNAi and eIF2 β overexpression and depletion). In the text, the authors simply state the observation of their LC3 blotting. The manuscript lacks an explanation of how to evaluate the LC3-II/LC3-I ratio. Also, the manuscript lacks an elaboration on what the results of the LC3 blotting indicate about the state of autophagy by the depletion of axonal mitochondria.

Thank you for pointing it out, and we apologize for an insufficient description of the result. We included quantitation of the levels of LC3-I and LC3-II in Figures 2A, 2D, 3D, 7B (Figure 6B in the previous version), and 8B (Figure 7B in the previous version). As the reviewer pointed out, LC3-II/LC3-I ratio changes do not necessarily indicate autophagy defects. However, since p62 accumulation (Figure 2B, 2E, 3E, 7C (Figure 6C in the previous version), 8C (Figure 7C in the previous version)), these results collectively suggest that autophagy is lowered. We revised the manuscript to include this discussion as below:

Lines 174-186 ‘During autophagy progression, LC3 is conjugated with phosphatidylethanolamine to form LC3-II, which localizes to isolation membranes and autophagosomes. LC3-I accumulation occurs when autophagosome formation is impaired, and LC3-II accumulation is associated with lysosomal defects^{31,32}. p62 is an autophagy substrate, and its accumulation suggests autophagic defects^{31,32}. We found that *milton* knockdown increased LC3-I, and the LC3-II/LC3-I ratio was lower in *milton* knockdown flies than in control flies at 14-day-old (Figure 2A). We also analyzed p62 levels in head lysates sequentially extracted using detergents with different stringencies (1% Triton X-100 and 2% SDS). Western blotting revealed that p62 levels were increased in the brains of 14-day-old of *milton* knockdown flies (Figure 2B). The increase in the p62 level was significant in the Triton X-100- soluble fraction but not in the SDS-soluble fraction (Figure 2B), suggesting that depletion of axonal mitochondria impairs the degradation of less-aggregated proteins.’

Line 189-190: ‘At 30 day-old, LC3-I was still higher, and the LC3-II/LC3-I ratio was lower, in *milton* knockdown compared to the control (Figure 2D).’

Line 202-203: ‘However, in contrast with *milton* knockdown, *Pfk* knockdown did not affect the levels of LC3-I, LC3-II or the LC3-II/LC3-I ratio (Figure 3D).’

Line 279-285: ‘Neuronal overexpression of *eIF2β* increased LC3-II, while the LC3-II/LC3-I ratio was not significantly different (Figure 7A and B). Overexpression of *eIF2β* significantly increased the p62 level in the Triton X-100-soluble fraction (Figure 7C, 4-fold vs. control, $p < 0.005$ (1% Triton X-100)) but not in the SDS-soluble fraction (Figure 7C, 2-fold vs. control, $p = 0.062$ (2% SDS)), as observed in brains of *milton* knockdown flies (Figure 2B). These data suggest that neuronal overexpression of *eIF2β* accumulates autophagic substrates.’

Line 311-319: ‘Neuronal knockdown of *milton* causes accumulation of autophagic substrate p62 in the Triton X-100-soluble fraction (Figure 2B), and we tested if lowering *eIF2β* ameliorates it. We found that *eIF2β* heterozygosity caused a mild increase in LC3-I levels and decreases in LC3-II levels, resulting in a significantly lower LC3-II/LC3-I ratio in *milton* knockdown flies (Figure 8B). *eIF2β* heterozygosity decreased the p62 level in the Triton X-100-soluble fraction in the brains of *milton* knockdown flies (Figure 8C). The p62 level in the SDS-soluble fraction, which is not sensitive to *milton* knockdown (Figure 2B), was not affected (Figure 8C). These results suggest that suppression of *eIF2β* ameliorates the impairment of autophagy caused by *milton* knockdown.’

Another main point of the paper is the up-regulation of *eIF2β* by depleting the axonal mitochondria leads to the proteostasis crisis. This claim is formed by the findings from the proteome analyses. The authors should have presented their proteomic data with much thorough presentation and explanation. As in the experiment scheme shown in Figure 4A, the author did two proteome analyses: one from the 7-day-old sample and the other from the 21-day-old sample. The manuscript only shows a plot of the result from the 7-day-old sample, but that of the result from the 21-day-old sample. For the 21-day-old sample, the authors only provided data in the supplemental table, in which the abundance ratio of *eIF2β* from the 21-day-old sample is 0.753, meaning *eIF2β* is depleted in the 21-day-old sample. The authors should have explained the impact of the

eIF2 β depletion in the 21-day-old sample, so the reader could fully understand the authors' interpretation of the role of eIF2 β on proteostasis.

NEW TEXT: Thank you for pointing it out. We included plots of the 21-day-old proteome results as a part of the main figure (Figure 4C). As the reviewer pointed out, *eIF2 β* protein levels are lower in *milton* knockdown background at the 21-day-old compared to the control. Since a reduction in the *eIF2 β* ameliorated *milton* knockdown-induced locomotor defects in aged flies (Figure 7D), the reduction in *eIF2 β* observed in the 21-day-old *milton* knockdown flies is not likely to negatively contribute to *milton* knockdown-induced defects. Our new data indicate that *eIF2 β* levels increase during aging in control flies until 49-day-old, then reduce at 63-day-old (included as Figure 4G in the revised manuscript). These age-dependent changes might explain the reduction in *eIF2 β* levels in *Milton* knockdown compared to the control in middle age: higher *eIF2 β* levels in *milton* knockdown flies at a young age than control and lower *eIF2 β* levels in the middle-aged flies may reflect premature aging.

NEW TEXT: We included these sentences in the discussion section:

NEW TEXT: Lines 240-243: 'We also investigated age-dependent changes in *eIF2 β* by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. *eIF2 β* levels increased during aging until 49-day-old (Figure 4G). These results suggest that upregulation of *eIF2 β* in *milton* knockdown fly brain reflects early an onset of age-dependent increase of *eIF2 β* levels.'

NEW TEXT: Lines 359-371: 'Our results suggest that the loss of axonal mitochondria is an event upstream of proteostasis collapse during aging. The number of puncta of ubiquitinated proteins was higher in *milton* knockdown at 14-day-old, but there was no significant difference at 30-day-old (Figure 1). Proteome analyses also showed that age-related pathways, such as immune responses, are enhanced in young flies with *milton* knockdown (Table 2). We also found that *eIF2 β* protein levels increase in an age-dependent manner until 49-day-old and reduces after that (Figure 4G). In the brains with neuronal knockdown of *milton*, *eIF2 β* levels were higher at 7-day-old than those in control and lower at the 21-day-old (Figure 4D and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude. Disruption of proteostasis is expected to contribute neurodegeneration³⁸, and it would be interesting to analyze the sequence of protein accumulation and axonal degeneration in *milton* knockdown (24,29 and Figure 1) in detail with higher time resolution.'

The manuscript consists of several weaknesses in its data and explanation regarding translation.

(1) The authors are likely misunderstanding the effect of phosphorylation of eIF2 α on translation. The P-eIF2 α is inhibitory for translation initiation. However, the authors seem to be mistaken that the down-regulation of P-eIF2 α inhibits translation.

We are sorry for our insufficient explanation in the previous version. As the reviewer pointed out, it is well known that the phosphorylated form of *eIF2 α* inhibits translation initiation. Neuronal knockdown of *milton* caused a reduction in p-*eIF2 α* (Figure 5D and E (Figure 4J and K in the previous version)), and it also lowered translation (Figure 6 (Figure 5 in the previous version)); the relationship between these two events is currently unclear. We do not think that a reduction in the p-*eIF2 α* suppressed translation; rather, we propose that the unbalance of expression levels of the components of *eIF2* complexes negatively affects translation. We revised discussion sections to describe our interpretation more in detail as below:

Line 374-384: '*eIF2 β* is a component of *eIF2*, which mediates translational regulation and ISR initiation. When ISR is activated, phosphorylated *eIF2 α* suppresses global translation and induces translation of ATF4, which mediates transcription of autophagy-related genes^{39,40}. Since ISR can positively regulate autophagy, we suspected that suppression of ISR underlies a

reduction in autophagic protein degradation. We found neuronal knockdown of *milton* reduced phosphorylated eIF2 α , suggesting that ISR is reduced (Figure 5). However, we also found that global translation was reduced (Figure 6). Increased levels of eIF2 β might disrupt the eIF2 complex or alter its functions. The stoichiometric mismatch caused by an imbalance of eIF2 components may inhibit ISR induction. Supporting this model, we found that eIF2 β upregulation reduced the levels of p-eIF2 α (Figure 7). We have revised the graphical abstract and removed the eIF2 complex since its role in the loss of proteostasis caused by *milton* knockdown has not been elucidated yet.

(2) The result of polysome profiling in Figure 4H is implausible. By 10%-25% sucrose density gradient, polysomes are not expected to be observed. The authors should have used a gradient with much denser sucrose, such as 10-50%.

Thank you for pointing it out. It was a mistake of 10-50%, and we apologize for the oversight. It was corrected (Figure 6 (Figure 5 in the previous version)).

(3) Also on the polysome profiling, as in the method section, the authors seemed to fractionate ultra-centrifuged samples from top to bottom and then measured A260 by a plate reader. In that case, the authors should have provided a line plot with individual data points, not the smoothly connected ones in the manuscript.

Thank you for pointing it out. We revised the graph (Figure 6 (Figure 5 in the previous version)).

(4) For both the results from polysome profiling and puromycin incorporation (Figure 4H and I), the difference between control siRNA and Milton siRNA are subtle, if not nonexistent. This might arise from the lack of spatial resolution in their experiment as the authors used head lysate for these data but the ratio of Phospho-eIF2 α /eIF2 α only changes in the axons, based on their results in Figure 4E-G. The authors could have attempted to capture the spatial resolution for the axonal translation to see the difference between control siRNA and Milton siRNA.

Thank you for your comment. We agree that it would be an interesting experiment, but it will take a considerable amount of time to analyze axonal translation with spatial resolution. We will try to include such analyses in the future. For this manuscript, we revised the discussion section to include the reviewer's suggestion as below;

Lines 355-357: 'Further analyses to dissect the effects of *milton* knockdown on proteostasis and translation in the cell body and axon by experiments with spatial resolution would be needed.'

Recommendations for the authors:

From the Reviewing Editor:

As the Reviewing Editor, I have read your manuscript and the associated peer reviews. I have concerns about publishing this work in its current form. I think that your manuscript cannot claim to have found a novel function of eIF2 β because of technical uncertainties and conceptual problems that should be addressed.

Thank you so much for your review and comments. We addressed all the concerns raised by the reviewers. Point-by-point responses are listed below.

First, your manuscript is based partly on what appears to be a mistaken understanding of the mechanistic basis of the ISR. Specifically, eIF2 is a heterotrimeric complex of alpha,

beta, and gamma subunits. When eIF2 α is phosphorylated, the heterotrimer adopts a new conformation. This conformation directly binds and inhibits eIF2B, the decameric GEF that exchanges the GDP bound to the gamma subunit of the eIF2 complex for GTP. Unless I misunderstood your paper, you seem to propose that decreasing levels of phospho-eIF2 α will inhibit translation, but this is backward from what we know about the ISR.

Thank you for your insightful comment, and we are sorry for the confusion. We did not mean to propose that decreasing levels of phospho-eIF2 α inhibits translation. We apologize for our insufficient explanation, which might have caused a misunderstanding (Lines 312-318 in the original version). We agree with the reviewer that ‘mismatch due to elevated eIF2-beta could change the behavior of the ISR’. We revised the text in the result section as follows:

Lines 263-268 (in the Result section) ‘Phosphorylation of eIF2 α induces conformational changes in the eIF2 complex and inhibits global translation³⁶. To analyze the effects of *milton* knockdown on translation, we performed polysome gradient centrifugation to examine the level of ribosome binding to mRNA. Since p-eIF2 α was downregulated, we hypothesized that *milton* knockdown would enhance translation. However, unexpectedly, we found that *milton* knockdown significantly reduced the level of mRNAs associated with polysomes (Figure 6A and B).’

Lines 374-384 (in the Discussion section): ‘eIF2 β is a component of eIF2, which mediates translational regulation and ISR initiation. When ISR is activated, phosphorylated eIF2 α suppresses global translation and induces translation of ATF4, which mediates transcription of autophagy-related genes^{39,40}. Since ISR can positively regulate autophagy, we suspected that suppression of ISR underlies a reduction in autophagic protein degradation. We found neuronal knockdown of *milton* reduced phosphorylated eIF2 α , suggesting that ISR is reduced (Figure 5). However, we also found that global translation was reduced (Figure 6). Increased levels of eIF2 β might disrupt the eIF2 complex or alter its functions. The stoichiometric mismatch caused by an imbalance of eIF2 components may inhibit ISR induction. Supporting this model, we found that eIF2 β upregulation reduced the levels of p-eIF2 α (Figure 7).’

It may be possible that a stoichiometric mismatch due to elevated eIF2-beta could change the behavior of the ISR, but your paper doesn't adequately address the expression levels of all three eIF2 subunits: alpha, beta, and gamma. The proteomic data shown in Fig 4B is unconvincing on its own because the changes in the beta subunit are subtle. The Western blot in Figure 4C suggests that the KD changes the mass or mobility of the beta subunit, and most importantly, there are no Western blots measuring the levels of eIF2 α , eIF2 α -phospho, or eIF2-gamma.

We appreciate the reviewer’s comment and agree that the stoichiometric mismatch due to elevated eIF2 β may interfere with ISR. We found overexpression of eIF2 β lowered p-eIF2 α (Figure S2 in V1), which supports this model. We included this data in the main figure in the revised manuscript (Figure 7D) and revised the text as below:

Lines 286-289: ‘Since *milton* knockdown reduced the p-eIF2 α level (Figure 5E), we asked whether an increase in eIF2 β affects p-eIF2 α . Neuronal overexpression of *eIF2 β* did not affect the eIF2 α level but significantly decreased the p-eIF2 α level (Figure 7D and E).’

Expression data of eIF2 α and eIF2 γ from proteomic analyses has been extracted from proteome analyses and included as a table (Figure 4D). Western blots of phospho-eIF2 α (Figure S1 in V1) in the main figure (Figure 5B). The result section was revised as below;

Lines 246-249: ‘As for the other subunits of eIF2 complex, proteome analysis did not detect a significant difference in the protein levels of eIF2 α and eIF2 γ between *milton* knockdown and control flies at 7 and 21 days (Figure 4D).’

NEW TEXT: We also analyzed age-dependent changes of eIF2 β by western blotting and found that eIF2 β increased during aging until 49-day-old. We included this result as Figure 4G and added these sentences in the result section:

NEW TEXT: Line 240-243: ‘We also investigated age-dependent changes in eIF2 β by western blotting of control flies at 7-, 21-, 35-, and 49-, and 63-day-old. eIF2 β levels increased during aging until 49-day-old (Figure 4G). These results suggest that upregulation of eIF2 β in *milton* knockdown fly brain reflects early an onset of age-dependent increase of eIF2 β levels.

Reviewer #1 (Recommendations For The Authors):

L125-128: In this section, while the efficiency of Milton knockdown is referenced from a previous publication, it is necessary to also mention that the Miro knockdown has been similarly reported in the literature. Additionally, the Methods section lacks details on the Miro RNAi line used, and Table 2 does not include the genotype for Miro RNAi. This information should be included for clarity and completeness.

Thank you for pointing it out. Knockdown efficiency with this strain has been reported (Iijima- Ando et al., PLoS Genet, 2012). We revised the text to include citation and knockdown efficiency as follows:

Lines 136-147: ‘There was no significant increase in ubiquitinated proteins in *milton* knockdown flies at 1-day old, suggesting that the accumulation of ubiquitinated proteins caused by *milton* knockdown is age-dependent (Figure S1). We also analyzed the effect of the neuronal knockdown of *Miro*, a partner of *milton*, on the accumulation of ubiquitin-positive proteins. Since severe knockdown of *Miro* in neurons causes lethality, we used UAS-*Miro* RNAi strain with low knockdown efficiency, whose expression driven by *elav*-GAL4 caused 30% reduction of *Miro* mRNA in head extract²⁴. Although there was a tendency for increased ubiquitin- positive puncta in *Miro* knockdown brains, the difference was not significant (Figure 1B, $p > 0.05$ between control RNAi and *Miro* RNAi). These data suggest that the depletion of axonal mitochondria induced by *milton* knockdown leads to the accumulation of ubiquitinated proteins before neurodegeneration occurs.’

L132-L136: The current phrasing in this section suggests an increase in ubiquitinated proteins for both Milton and Miro knockdowns. However, since there is no significant difference noted for Miro, it is incorrect to state an increase in ubiquitin-positive puncta. Furthermore, combining the results of Milton knockdown to claim an increase in ubiquitinated proteins prior to neurodegeneration is misleading. At the very least, the expression here needs to be moderated to accurately reflect the findings.

Thank you for pointing it out. We revised the text as above.

L137-L141: Results in Figure 1 indicate that Milton knockdown leads to an increase in ubiquitinated proteins at 14 days, while Miro knockdown shows no difference from the control at either 14 or 30 days. Conversely, both the control and Miro exhibit an increase in ubiquitinated proteins with aging, but this trend does not seem to apply to Milton knockdown. This observation suggests that Milton KD may not affect the changes in protein quality control associated with aging. It implies that Milton's function might be more related to protein homeostasis in younger cells, or that changes due to aging might overshadow the effects of Milton knockdown. These interpretations should be included in the Results or Discussion sections for a more comprehensive analysis.

NEW TEXT: Thank you for your insightful comment. As you mentioned, the accumulation of ubiquitinated proteins significantly increases only in young flies. Age-related pathways, such

as immune responses, are highlighted in young *milton* knockdown flies but not in the aged flies. Our new result indicates that eIF2 β increases during aging in control flies (included as Figure 4G in the revised manuscript), and upregulation of eIF2 β in *milton* knockdown is only observed at a young age. These results suggest that *milton* knockdown does not increase the magnitude of age-dependent changes but accelerates their onset. We revised the text to include those points as follows:

NEW TEXT: Lines 152-153: ‘These results suggest that depletion of axonal mitochondria may have more impact on proteostasis in young neurons than in old neurons.’

NEW TEXT: Lines 359-371: ‘Our results suggest that the loss of axonal mitochondria is an event upstream of proteostasis collapse during aging. The number of puncta of ubiquitinated proteins was higher in *milton* knockdown at 14-day-old, but there was no significant difference at 30-day-old (Figure 1). Proteome analyses also showed that age-related pathways, such as immune responses, are enhanced in young flies with *milton* knockdown (Table 2). We also found that eIF2 β protein levels increase in an age-dependent manner until 49-day-old and reduces after that (Figure 4G). In the brains with neuronal knockdown of *milton*, eIF2 β levels were higher at 7-day-old than those in control and lower at the 21-day-old (Figure 4 and Supplementary table). These results suggest that *milton* knockdown is likely accelerating age-dependent changes rather than increasing their magnitude. Disruption of proteostasis is expected to contribute neurodegeneration³⁸, and it would be interesting to analyze the sequence of protein accumulation and axonal degeneration in *milton* knockdown (24,29 and Figure 1) in detail with higher time resolution.’

L143 : Please remove the erroneously included quotation mark.

Thank you for pointing it out. We corrected it.

L145-L147:

While it is understood that Milton knockdown results in a reduction of mitochondria in axons, as reported previously and seemingly indicated in Figure 1E, this paper repeatedly refers to axonal depletion of mitochondria. Therefore, it would be beneficial to quantitatively assess the number of mitochondria in the axonal terminals located in the lamina via electron microscopy. Such quantification would robustly reinforce the argument that mitochondrial absence in axons is a consequence of Milton knockdown.

Thank you for pointing it out. We included quantitation of the number of mitochondria in the synaptic terminals (Figure 1E).

The text and figure legend was revised accordingly:

Lines 156-157: ‘As previously reported²⁴, the number of mitochondria in presynaptic terminals decreased in *milton* knockdown (Figure 1E).’

The knockdown of Milton is known to reduce mitochondrial transport from an early stage, but what about swelling? By observing swelling at 1 day and 14 days, it may be possible to confirm the onset of swelling and discuss its correlation with the accumulation of ubiquitinated proteins.

Quantitation of axonal swelling has also been included (Figure 1F).

We appreciate the reviewer's comments on the correlation between the accumulation of ubiquitinated proteins and axonal swelling. Axonal swelling was not observed at 3-days-old (Iijima-Ando et al., PLoS Genetics, 2012), indicating that axonal swelling is an age-dependent event. Dense materials are found in swollen axons more often than in normal axons,

suggesting a positive correlation between disruption of proteostasis and axonal damage. It would be interesting to analyze the time course of events further; however, we feel it is beyond the scope of this manuscript. We revised the text to include this discussion as:

Lines 157-160: ‘The swelling of presynaptic terminals, characterized by the enlargement and roundness, was not reported at 3-day-old²⁴ but observed at this age with about 4% of total presynaptic terminals (Figure 1F, asterisks).’

Lines 162-167: ‘Dense materials are rarely found in age-matched control neurons, indicating that *milton* knockdown induces abnormal protein accumulation in the presynaptic terminals (Figure 1G and H). In *milton* knockdown neurons, dense materials are found in swollen presynaptic terminals more often than in presynaptic terminals without swelling, suggesting a positive correlation between the disruption of proteostasis and axonal damage (Figure 1G).’

Lines 369-371: ‘Disruption of proteostasis is expected to contribute neurodegeneration³⁸, and it would be interesting to analyze the sequence of protein accumulation and axonal degeneration in *milton* knockdown (24,29 and Figure 1) in detail with higher time resolution.’

L147-L151: Though Figures 1F and 1G provide qualitative representations, it is advisable to quantitatively assess whether dense materials significantly accumulate. Such quantitative analysis would be required to verify the accumulation of dense materials in the context of the study.

Thank you for pointing it out. We included quantitation of the number of neurons with dense material (Figure 1G). We revised the manuscript as follows:

Line 162-164: ‘Dense materials are rarely found in age-matched control neurons, indicating that *milton* knockdown induces abnormal protein accumulation in the presynaptic terminals (Figure 1G and H).’

Regarding Figure 1B, C:

Even though the count of puncta in the whole brain appears to be fewer than 400, the magnification of the optic lobe suggests a substantial presence of puncta. Please clarify in the Methods section what constitutes a puncta and whether the quantification in the whole brain is based on a 2D or 3D analysis. Detail the methodology used for quantification.

Thank you for your comment. We revised the method section to include more details as below:

Lines 440-443: ‘Quantitative analysis was performed using ImageJ (National Institutes of Health) with maximum projection images derived from Z-stack images acquired with same settings. Puncta was identified with mean intensity and area using ImageJ.’

What about 1-day-old specimens? Does Milton knockdown already show an increase in ubiquitinated protein accumulation at this early stage? Investigating whether ubiquitin-protein accumulation is involved in aging promotion or is already prevalent during developmental stages is a necessary experiment.

Thank you for your comment. We carried out immunostaining with an anti-ubiquitin antibody in the brains at 1-day-old. No significant difference was detected between the control and *milton* knockdown. This result has been included as Figure S1 in the revised manuscript. The result section was revised as below:

Line 136-139 ‘There was no significant increase in ubiquitinated proteins in *milton* knockdown flies at 1-day old, suggesting that the accumulation of ubiquitinated proteins caused by *milton* knockdown is age-dependent (Figure S1).’

For Figure 1E: In the Electron Microscopy section of the Methods, define how swollen axons were identified and describe the quantification methodology used.

Thank you for your comment. Swollen axons are, unlike normal axons, round in shape and enlarged. We revised the text as below;

Lines 157-160: ‘The swelling of presynaptic terminals, characterized by the enlargement and roundness, was not reported at 3-day-old²⁴ but observed at this age with about 4% of total presynaptic terminals (Figure 1F, asterisks).’

Lines 689-691, Figure 1 legend: ‘Swollen presynaptic terminals (asterisks in (F)), characterized by the enlargement and higher circularity, were found more frequently in *milton* knockdown neurons.’

L218-L219: Throughout the text, the expression 'eIF2 β is "upregulated" in response to Milton knockdown' is frequently used. However, considering the presented results, it might be more accurate to interpret that under the condition of Milton knockdown, eIF2 β is not undergoing degradation but rather remains stable.

Thank you for pointing it out. We replaced ‘upregulated’ with ‘increased’ throughout the text.

L234-L235: On what basis is the conclusion drawn that there is a reduction? Given that three experiments have been conducted, it would be possible and more convincing to quantify the results to determine if there is a significant decrease.

Thank you for pointing it out. We quantified the AUC of polysome fraction and carried out a statistical analysis. There is a significant decrease in polysome in *milton* knockdown, and this result has been included in Figure 5B. We revised the figure and the legend accordingly.

L236: 5H-> 4H

Thank you for pointing it out, and we are sorry for the confusion. We corrected it.

L238-L239: Since there is no significant difference observed, it may not be accurate to interpret a reduction in puromycin incorporation.

Thank you for pointing it out. As described above, quantification of polysome fractions showed that *milton* knockdown significantly reduced polysome (Figure 6B (Figure 5B in the previous version)). We revised the manuscript as below;

Lines 267-268: ‘However, unexpectedly, we found that *milton* knockdown significantly reduced the level of mRNAs associated with polysomes (Figure 6A and B).’

Figure 5D and Figure 6D: Climbing assays have been conducted, but I believe experiments should also be performed to examine whether overexpression or heterozygous mutants of eIF2 β induce or suppress degeneration.

Thank you for pointing it out. We analyzed the eyes with *eIF2 β* overexpression for neurodegeneration. Although there was a tendency of elevated neurodegeneration in the retina with *eIF2 β* overexpression, the difference between control and *eIF2 β* overexpression

did not reach statistical significance (Figure S2). This result has been included as Figure S2 in the revised manuscript, and the following sentences have been included in the text:

Lines 292-297: ‘We asked if *eIF2 β* overexpression causes neurodegeneration, as depletion of axonal mitochondria in the photoreceptor neurons causes axon degeneration in an age-dependent manner²⁴. *eIF2 β* overexpression in photoreceptor neurons tends to increase neurodegeneration in aged flies, while it was not statistically significant ($p > 0.05$, Figure S2).’

L271-L272: The results in Figure 6B are surprising. I anticipated a greater increase compared to the Milton knockdown alone. While p62 appears to be reduced, it is not clear why these results lead to the conclusion that lowering eIF2 β rescues autophagic impairment. Please add a discussion section to address this point.

Thank you for pointing it out. We apologize for the unclear description of the result. Milton knockdown flies show p62 accumulation (Figure 2), and deleting one copy of *eIF2beta* in *milton* knockdown background reduced p62 accumulation (Figure 8C (Figure 7C in the previous version)). We revised the text as below:

Lines 311-319: ‘Neuronal knockdown of *milton* causes accumulation of autophagic substrate p62 in the Triton X-100-soluble fraction (Figure 2B), and we tested if lowering *eIF2 β* ameliorates it. We found that *eIF2 β* heterozygosity caused a mild increase in LC3-I levels and decreases in LC3-II levels, resulting in a significantly lower LC3-II/LC3-I ratio in *milton* knockdown flies (Figure 8B). *eIF2 β* heterozygosity decreased the p62 level in the Triton X-100-soluble fraction in the brains of *milton* knockdown flies (Figure 8C). The p62 level in the SDS-soluble fraction, which is not sensitive to *milton* knockdown (Figure 2B), was not affected (Figure 8C). These results suggest that suppression of *eIF2 β* ameliorates the impairment of autophagy caused by *milton* knockdown.’

L369: Please specify the source of the anti-ubiquitin antibody used.

Thank you for pointing it out. We included the antibody information in the method section.

*Figure 7: While the relationship between Milton knockdown and the eIF2 β and eIF2 α proteins has been elucidated through the authors' efforts, I would like to see an investigation into whether eIF2 β is upregulated and eIF2 α phosphorylation is reduced in simply aged *Drosophila*. This would help us understand the correlation between aging and eIF2 protein dynamics.*

Thank you for your comment. We agree that it is an important question, and we are working on it. However, we feel that it is beyond the scope of the current manuscript.

L645-L646: If the mushroom body is identified using mito-GFP, then include mito-GFP in the genotype listed in Supplementary Table 2.

We are sorry for the oversight. We corrected it in Supplementary Table 2.

Additionally, while it is presumed that the mito-GFP signal decreases in axons with Milton RNAi, how was the lobe tips area accurately selected for analysis? Please include these details along with a comprehensive description of the quantification methodology in the Methods section.

Thank you for your comment. Although the mito-GFP signal in the axon is weak in the *milton* knockdown neurons, it is sufficient to distinguish the mushroom body structure from the background. We revised the method section to include this information in the method section:

Line 443-447: ‘For eIF2 α and p-eIF2 α immunostaining, the mushroom body was detected by mitoGFP expression.’

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